#1 · General Pathology

What distinguishes hypoxia from ischaemia?

Answer

Hypoxia is reduced oxygen supply. Ischaemia is reduced blood supply, which also removes nutrients and impairs washout of metabolites, so it injures tissue faster.

#2 · General Pathology

What is the earliest biochemical event in hypoxic cell injury?

Answer

Loss of oxidative phosphorylation with a fall in ATP. This fails the sodium-potassium pump, so sodium and water enter the cell and it swells.

#3 · General Pathology

What are the two microscopic hallmarks of reversible cell injury?

Answer

Cellular swelling (hydropic change) and fatty change. Both can be reversed if the insult is removed.

#4 · General Pathology

Which imaging findings indicate hepatic steatosis on ultrasound, CT and MRI?

Answer

Increased echogenicity on ultrasound, reduced attenuation relative to the spleen on unenhanced CT, and signal loss on out-of-phase chemical shift MRI.

#5 · General Pathology

What phenomena consistently mark the point of irreversible cell injury?

Answer

Inability to reverse mitochondrial dysfunction after reperfusion, and profound plasma membrane damage.

#6 · General Pathology

Name the three nuclear changes of necrosis.

Answer

Pyknosis (shrinkage and hyperchromasia), karyorrhexis (fragmentation) and karyolysis (fading of the nucleus).

#7 · General Pathology

Which pattern of necrosis typically follows ischaemia in solid organs other than brain, and what is its imaging correlate?

Answer

Coagulative necrosis, with preserved architecture. In kidney or spleen it appears as a peripheral wedge of non-enhancement with the base on the capsule.

#8 · General Pathology

Which tissues undergo liquefactive necrosis?

Answer

Bacterial abscesses and cerebral infarcts. The tissue is digested by hydrolytic enzymes and becomes a cyst-like space.

#9 · General Pathology

What are the macroscopic and microscopic features of caseous necrosis?

Answer

Macroscopically soft, white and cheese-like. Microscopically amorphous granular eosinophilic debris with no preserved architecture, ringed by a granulomatous reaction.

#10 · General Pathology

What is saponification and where does it occur?

Answer

Fatty acids released by lipases combine with calcium to form soaps in fat necrosis, as in acute pancreatitis. Macroscopically these are chalky white areas.

#11 · General Pathology

In which conditions is fibrinoid necrosis seen?

Answer

Immune-mediated vasculitis and malignant hypertension, where immune complexes and fibrin deposit in vessel walls.

#12 · General Pathology

What is the imaging sign of gas gangrene?

Answer

Gas locules within soft tissue on radiograph or CT, due to Clostridium species.

#13 · General Pathology

When do infarcts appear red rather than white?

Answer

With venous occlusion, in loose tissue such as lung, in organs with dual circulation, and when blood reflows into necrotic tissue. White infarcts occur in solid organs with end-arterial supply.

#14 · General Pathology

How does the apoptotic cell differ from the necrotic cell microscopically?

Answer

Apoptosis shows a shrunken cell with dense eosinophilic cytoplasm, condensed fragmented chromatin and apoptotic bodies, with an intact membrane and no inflammation.

#15 · General Pathology

How do phagocytes recognise apoptotic cells?

Answer

By phosphatidylserine flipped to the outer leaflet of the plasma membrane.

#16 · General Pathology

Which proteins regulate the intrinsic apoptotic pathway?

Answer

Anti-apoptotic BCL2 and BCL-XL oppose pro-apoptotic BAX and BAK. When BAX and BAK dominate, cytochrome c is released and forms the apoptosome with APAF-1 to activate caspase 9.

#17 · General Pathology

Which caspase initiates the extrinsic (death receptor) pathway?

Answer

Caspase 8, activated by FAS ligand binding FAS or by TNF binding its receptor. Both pathways converge on executioner caspases 3 and 6.

#18 · General Pathology

Which translocation links apoptosis failure to follicular lymphoma?

Answer

t(14;18), which overexpresses BCL2 and blocks apoptosis.

#19 · General Pathology

What is the role of p53 in cell injury?

Answer

p53 induces cell cycle arrest and, if DNA damage cannot be repaired, apoptosis.

#20 · General Pathology

Distinguish dystrophic from metastatic calcification.

Answer

Dystrophic calcification occurs in dead or damaged tissue with normal serum calcium. Metastatic calcification occurs in normal tissue and always reflects hypercalcaemia.

#21 · General Pathology

Which tissues favour metastatic calcification?

Answer

Tissues with an alkaline interior because they lose acid: gastric mucosa, kidney, lung and systemic arteries.

#22 · General Pathology

Which lesions classically show psammomatous calcification?

Answer

Papillary thyroid carcinoma, meningioma and serous ovarian tumours.

#23 · General Pathology

What conditions are associated with eggshell nodal calcification?

Answer

Silicosis and sarcoidosis.

#24 · General Pathology

How does haemosiderin appear, and what is the MRI correlate of iron overload in the liver?

Answer

Golden-brown pigment staining blue with Prussian blue. The liver shows marked signal loss on T2-weighted and gradient-echo MRI.

#25 · General Pathology

What are the five classic signs of acute inflammation?

Answer

Rubor, tumor, calor, dolor and functio laesa (redness, swelling, heat, pain and loss of function).

#26 · General Pathology

How does an exudate differ from a transudate?

Answer

An exudate is protein-rich and results from increased vascular permeability. A transudate is protein-poor and results from hydrostatic or osmotic imbalance.

#27 · General Pathology

List the sequence of leukocyte recruitment.

Answer

Margination, rolling, adhesion, transmigration and chemotaxis.

#28 · General Pathology

Which molecules mediate rolling and firm adhesion of leukocytes?

Answer

Selectins (P, E and L) mediate rolling. Integrins binding ICAM-1 and VCAM-1 mediate firm adhesion, and PECAM-1 mediates transmigration.

#29 · General Pathology

Which cell predominates early and which later in acute inflammation?

Answer

Neutrophils in the first 6 to 24 hours, then monocytes and macrophages at 24 to 48 hours.

#30 · General Pathology

What is the defect in chronic granulomatous disease?

Answer

Deficiency of NADPH oxidase, so reactive oxygen species are not generated and catalase-positive organisms cause recurrent infection.

#31 · General Pathology

What is the defect in leukocyte adhesion deficiency?

Answer

Defective integrins, causing delayed umbilical cord separation and recurrent infection without pus.

#32 · General Pathology

Which mediator is the potent complement chemotactic agent?

Answer

C5a.

#33 · General Pathology

What is the imaging correlate of an abscess?

Answer

A rim-enhancing collection with a central region of restricted diffusion on MRI, with gas locules if gas-forming organisms are present.

#34 · General Pathology

What is the dominant cell in chronic inflammation and which cytokine activates the M1 type?

Answer

The macrophage. Interferon gamma from T helper 1 cells produces the classically activated M1 macrophage, which kills microbes.

#35 · General Pathology

What is a granuloma and how do immune and foreign body types differ?

Answer

A focal aggregate of epithelioid macrophages, often with giant cells. Immune granulomas follow a T cell response to poorly degradable antigen. Foreign body granulomas arise from inert material without a T cell response.

#36 · General Pathology

Which diseases produce non-caseating granulomas?

Answer

Sarcoidosis, Crohn disease and berylliosis. Caseating granulomas are characteristic of tuberculosis.

#37 · General Pathology

Distinguish a Ghon complex from a Ranke complex.

Answer

A Ghon complex is the subpleural Ghon focus plus the enlarged hilar node. When it calcifies it is called a Ranke complex.

#38 · General Pathology

What is the difference between a hypertrophic scar and a keloid?

Answer

A hypertrophic scar stays within the wound margin. A keloid extends beyond the original wound and has thick collagen bundles.

#39 · General Pathology

Which tissue classes can regenerate and which heal by scarring?

Answer

Labile and stable tissues regenerate. Permanent tissues such as neurons and cardiac myocytes heal by scarring.

#40 · General Pathology

What are the features of granulation tissue?

Answer

Macroscopically pink, soft and granular. Microscopically proliferating fibroblasts and new capillaries in loose matrix, with angiogenesis driven by VEGF.

#41 · General Pathology

What are the Congo red findings and main amyloid types?

Answer

Apple-green birefringence under polarised light. AL arises from light chains, AA from serum amyloid A, ATTR from transthyretin, and beta 2 microglobulin occurs in dialysis patients.

#42 · General Pathology

How does bone scintigraphy help distinguish ATTR from AL cardiac amyloid?

Answer

ATTR shows intense myocardial uptake of technetium pyrophosphate or diphosphonate tracer. AL is typically negative or weak.

#43 · General Pathology

What is the difference between dysplasia and carcinoma in situ?

Answer

Dysplasia is disordered epithelial growth that has not breached the basement membrane. Carcinoma in situ is full thickness dysplasia, still without invasion.

#44 · General Pathology

Which imaging features favour malignancy over benign disease?

Answer

Irregular or spiculated margin, invasion of fat planes, vessels or bone, necrosis and rapid growth. These are probabilities and histology decides.

#45 · General Pathology

What are the three routes of tumour spread?

Answer

Seeding of body cavities (transcoelomic), lymphatic spread and haematogenous spread.

#46 · General Pathology

Which route of spread is the usual early route for carcinomas and which is favoured by sarcomas?

Answer

Carcinomas usually spread first by lymphatics. Sarcomas favour haematogenous spread.

#47 · General Pathology

Why is the liver the first destination for colorectal metastases?

Answer

Gut venous blood drains through the portal vein to the liver.

#48 · General Pathology

Which two tumours characteristically grow as tumour thrombus in veins?

Answer

Renal cell carcinoma into the renal vein and inferior vena cava, and hepatocellular carcinoma into the portal and hepatic veins. Tumour thrombus enhances after contrast, unlike bland thrombus.

#49 · General Pathology

What early change allows tumour cells to detach from each other?

Answer

Loss of E cadherin.

#50 · General Pathology

How do grading and staging differ?

Answer

Grading is histological estimate of aggressiveness by differentiation and mitotic count. Staging is the extent of spread, and it usually carries more prognostic weight.

#51 · General Pathology

What do the prefixes c, p and y mean in TNM staging?

Answer

c is clinical staging including imaging, p is pathological staging after resection, y is staging after neoadjuvant therapy.

#52 · General Pathology

Which is the most commonly mutated oncogene in human tumours, and which is the most commonly mutated tumour suppressor?

Answer

RAS is the most commonly mutated oncogene. TP53 is the most commonly mutated tumour suppressor gene.

#53 · General Pathology

What translocation activates MYC in Burkitt lymphoma?

Answer

t(8;14).

#54 · General Pathology

Which translocation produces the BCR-ABL fusion and in which disease?

Answer

The Philadelphia chromosome t(9;22), in chronic myeloid leukaemia.

#55 · General Pathology

What is the prognostic significance of N-MYC amplification?

Answer

It occurs in neuroblastoma and indicates poor prognosis.

#56 · General Pathology

State Knudson's two hit hypothesis.

Answer

Both alleles of a tumour suppressor gene must be lost. In hereditary cases the first hit is inherited in every cell, so tumours arise earlier and are often bilateral or multifocal.

#57 · General Pathology

What is the Warburg effect and its imaging relevance?

Answer

Tumours take up glucose and use glycolysis even when oxygen is available. This is the basis of FDG PET.

#58 · General Pathology

Which tumour causes hypercalcaemia through PTHrP?

Answer

Typically squamous cell carcinoma of lung.

#59 · General Pathology

Which paraneoplastic syndromes are linked to small cell lung carcinoma?

Answer

ACTH secretion causing Cushing syndrome, ADH secretion causing SIADH, and Lambert Eaton myasthenic syndrome.

#60 · General Pathology

Which tumours can cause polycythaemia through erythropoietin?

Answer

Renal cell carcinoma, hepatocellular carcinoma and cerebellar haemangioblastoma.

#61 · General Pathology

What is Trousseau syndrome and which tumour is classically associated?

Answer

Migratory thrombophlebitis, classically with pancreatic carcinoma.

#62 · General Pathology

What are the key features of neurofibromatosis type 1?

Answer

NF1 loss on chromosome 17: neurofibromas including plexiform, café au lait macules, Lisch nodules, optic pathway glioma and sphenoid wing dysplasia.

#63 · General Pathology

Which tumours define neurofibromatosis type 2?

Answer

Bilateral vestibular schwannomas, with meningiomas and ependymomas. The gene is NF2 on chromosome 22.

#64 · General Pathology

Which renal lesion occurs in tuberous sclerosis and what is its imaging key?

Answer

Angiomyolipoma, which contains macroscopic fat.

#65 · General Pathology

Name the CNS lesions of tuberous sclerosis.

Answer

Cortical tubers, subependymal nodules and subependymal giant cell astrocytoma near the foramen of Monro.

#66 · General Pathology

Which tumours characterise von Hippel Lindau disease?

Answer

Haemangioblastomas of cerebellum, spinal cord and retina, clear cell renal carcinoma, phaeochromocytoma, and pancreatic cysts or neuroendocrine tumours.

#67 · General Pathology

What gene defect causes Marfan syndrome and what are its cardiovascular risks?

Answer

Fibrillin 1 defect, causing aortic root dilatation and dissection.

#68 · General Pathology

What is the pulmonary pattern of cystic fibrosis on imaging?

Answer

Bronchiectasis predominantly in the upper lobes with mucus plugging. The gene is CFTR on chromosome 7.

#69 · General Pathology

What is the molecular change in sickle cell disease and a skeletal consequence?

Answer

Valine replaces glutamic acid at position 6 of beta globin. Bone infarcts and avascular necrosis of the femoral head result.

#70 · General Pathology

Which abnormalities are associated with Down syndrome?

Answer

Atrioventricular septal defect, duodenal atresia (double bubble sign), Hirschsprung disease, atlantoaxial instability and increased risk of acute leukaemia.

#71 · General Pathology

Which cardiovascular and renal anomalies occur in Turner syndrome (45,X)?

Answer

Coarctation of the aorta, bicuspid aortic valve and horseshoe kidney, with streak ovaries.

#72 · General Pathology

What tumour risks accompany Klinefelter syndrome (47,XXY)?

Answer

Male breast cancer and mediastinal germ cell tumours.

#73 · General Pathology

Which Starling force changes explain oedema in nephrotic syndrome and in heart failure?

Answer

Nephrotic syndrome lowers plasma oncotic pressure through hypoalbuminaemia. Heart failure raises capillary hydrostatic pressure.

#74 · General Pathology

How do transudate and exudate differ, and what does an exudate indicate?

Answer

A transudate is protein poor with low specific gravity. An exudate is protein rich and indicates inflammation or malignancy through increased permeability.

#75 · General Pathology

How does vasogenic cerebral oedema differ from cytotoxic oedema?

Answer

Vasogenic oedema follows blood brain barrier breakdown, spreads through white matter and spares grey matter. Cytotoxic oedema follows cell swelling in infarction and involves both grey and white matter.

#76 · General Pathology

What are the three components of Virchow's triad?

Answer

Endothelial injury, abnormal blood flow (stasis or turbulence) and hypercoagulability.

#77 · General Pathology

What is the most common inherited hypercoagulable state and its mechanism?

Answer

Factor V Leiden, where factor V is resistant to inactivation by activated protein C.

#78 · General Pathology

What do lines of Zahn indicate?

Answer

Alternating pale platelet and fibrin layers and darker red cell layers, distinguishing antemortem thrombus from postmortem clot.

#79 · General Pathology

What is the CT sign of acute pulmonary embolism and of right heart strain?

Answer

A central filling defect surrounded by contrast. Right heart strain shows a dilated right ventricle, right to left ventricular ratio above 1, septal bowing to the left and contrast reflux into the inferior vena cava.

#80 · General Pathology

When does fat embolism syndrome present and what is the MRI pattern?

Answer

One to three days after long bone fracture or trauma, with respiratory failure, neurological signs and petechial rash. Brain MRI shows a starfield pattern of small foci of restricted diffusion.

#81 · General Pathology

Which infarcts are white and which are red?

Answer

White infarcts follow arterial occlusion in solid organs such as heart, spleen and kidney. Red infarcts follow venous occlusion or occur in loose or dual supply tissue such as lung and bowel.

#82 · General Pathology

What are the CT features of the hypoperfusion complex in shock?

Answer

Flat inferior vena cava, small aorta, diffuse bowel wall thickening with intense mucosal enhancement and increased adrenal enhancement.

#83 · General Pathology

Name the four types of hypersensitivity with one example each.

Answer

Type I IgE anaphylaxis, type II antibody against cell antigens as in Goodpasture syndrome, type III immune complex as in lupus, type IV T cell mediated as in tuberculosis.

#84 · General Pathology

Why is pulmonary infarction uncommon after pulmonary embolism, and what does it look like?

Answer

The lung has dual bronchial and pulmonary supply. When it occurs it is a peripheral pleural based wedge shaped opacity, the Hampton hump.

#85 · Cardiovascular and Vascular

Which cell type and matrix make up the fibrous cap of an atheromatous plaque, and what features mark a vulnerable plaque?

Answer

The cap is smooth muscle cells, collagen and macrophages. A vulnerable plaque has a thin cap, a large lipid core and many macrophages; a stable plaque has a thick collagenous cap, often with more calcium.

#86 · Cardiovascular and Vascular

What is the order of severity of atherosclerosis by vessel?

Answer

Infrarenal abdominal aorta, then proximal coronary arteries (LAD first), popliteal arteries, internal carotid arteries and the circle of Willis.

#87 · Cardiovascular and Vascular

How does Monckeberg medial calcific sclerosis differ from atherosclerosis?

Answer

It calcifies the media of muscular arteries, does not narrow the lumen and is common in the elderly and in diabetes. Atherosclerosis is an intimal plaque disease.

#88 · Cardiovascular and Vascular

What cell is the hallmark of the fatty streak, and is it clinically significant?

Answer

The macrophage-derived foam cell. The fatty streak is a flat yellow lesion with no flow limitation and is not clinically important.

#89 · Cardiovascular and Vascular

What are the complications of an atherosclerotic plaque?

Answer

Rupture or erosion with thrombosis, haemorrhage into the plaque, atheroembolism of cholesterol crystals, and aneurysm formation through medial atrophy.

#90 · Cardiovascular and Vascular

Which coronary artery most often causes myocardial infarction, and what does it supply?

Answer

The left anterior descending artery (about 40 to 50 percent). It supplies the anterior left ventricle, the anterior two thirds of the septum and the apex.

#91 · Cardiovascular and Vascular

Which coronary territory does the right coronary artery supply?

Answer

The posterior left ventricular wall, the posterior septum and the right ventricle; it is the dominant vessel in most people.

#92 · Cardiovascular and Vascular

What are the light microscopic findings in an infarct at 1 to 3 days?

Answer

Coagulative necrosis with loss of nuclei and striations and a maximal neutrophil infiltrate; the infarct is yellow-tan and mottled grossly.

#93 · Cardiovascular and Vascular

When is the myocardium at greatest risk of rupture after infarction, and why?

Answer

Days 3 to 7, when macrophages have removed necrotic muscle and the area is at maximal softening and has not yet been replaced by collagen.

#94 · Cardiovascular and Vascular

When does a myocardial infarct become a mature white scar?

Answer

By about two months, after granulation tissue (7 to 10 days) is replaced by collagen deposition over weeks 2 to 8.

#95 · Cardiovascular and Vascular

Why is the posteromedial papillary muscle more often ruptured than the anterolateral?

Answer

It has a single blood supply from the posterior descending artery, whereas the anterolateral muscle has a dual supply.

#96 · Cardiovascular and Vascular

Which complication of infarction is the commonest cause of death in the first hours?

Answer

Arrhythmia, especially ventricular fibrillation.

#97 · Cardiovascular and Vascular

What distinguishes a true ventricular aneurysm from a pseudoaneurysm after infarction?

Answer

A true aneurysm is a scarred bulge containing myocardium replaced by scar and rarely ruptures. A pseudoaneurysm is a contained rupture walled by pericardium and thrombus and can rupture.

#98 · Cardiovascular and Vascular

What pattern of late gadolinium enhancement indicates ischaemic scar on cardiac MRI?

Answer

Subendocardial enhancement extending towards the epicardium in a coronary territory, in contrast to midwall or epicardial enhancement in non-ischaemic disease.

#99 · Cardiovascular and Vascular

Which type of infarct follows global hypotension, and where does it lie?

Answer

A subendocardial infarct, confined to the inner third to half of the wall, the zone most vulnerable to hypoperfusion.

#100 · Cardiovascular and Vascular

What is the commonest genetic cause of dilated cardiomyopathy?

Answer

Truncating variants in the titin gene (TTN). Other causes include alcohol, myocarditis, peripartum disease, anthracyclines and haemochromatosis.

#101 · Cardiovascular and Vascular

Describe the macroscopic and microscopic hallmarks of hypertrophic cardiomyopathy.

Answer

Asymmetric septal hypertrophy with a small left ventricular cavity; microscopically myocyte hypertrophy, myofibre disarray and interstitial fibrosis.

#102 · Cardiovascular and Vascular

What is the inheritance and typical gene involvement in hypertrophic cardiomyopathy?

Answer

Autosomal dominant, with mutations in sarcomere genes such as MYH7 and MYBPC3.

#103 · Cardiovascular and Vascular

How do restrictive cardiomyopathies differ morphologically from dilated forms?

Answer

Ventricles are normal or near normal in size with stiff walls and often biatrial dilation. Causes include amyloid, sarcoid, haemochromatosis and radiation.

#104 · Cardiovascular and Vascular

What staining feature identifies cardiac amyloid?

Answer

Congo red staining with apple-green birefringence under polarised light; the main forms are transthyretin and light chain.

#105 · Cardiovascular and Vascular

What defines Takotsubo cardiomyopathy?

Answer

Transient apical ballooning with basal hyperkinesis after severe stress, mediated by catecholamines, with normal coronary arteries.

#106 · Cardiovascular and Vascular

What is the pathological change in arrhythmogenic right ventricular cardiomyopathy?

Answer

Replacement of right ventricular myocardium by fat and fibrous tissue, usually from desmosomal gene mutations, causing arrhythmia and sudden death.

#107 · Cardiovascular and Vascular

What is the pathognomonic lesion of rheumatic heart disease?

Answer

The Aschoff body, a granuloma with fibrinoid necrosis and Anitschkow cells with caterpillar-shaped nuclei.

#108 · Cardiovascular and Vascular

What is the mechanism of rheumatic fever and the usual chronic valve lesion?

Answer

Molecular mimicry after group A streptococcal pharyngitis; the chronic lesion is mitral stenosis with commissural fusion and a fish-mouth orifice.

#109 · Cardiovascular and Vascular

How does calcific aortic stenosis differ from rheumatic aortic stenosis?

Answer

Calcific disease has calcium on the aortic side of the cusps without commissural fusion; rheumatic disease fuses the commissures.

#110 · Cardiovascular and Vascular

At what age does a bicuspid aortic valve typically become stenotic?

Answer

The fifth or sixth decade, earlier than the seventh or eighth decade of a normal tricuspid valve.

#111 · Cardiovascular and Vascular

Which valves are affected in carcinoid heart disease, and why?

Answer

The tricuspid and pulmonary valves, because the lungs inactivate serotonin released from hepatic metastases.

#112 · Cardiovascular and Vascular

What is the typical appearance of non-bacterial thrombotic endocarditis?

Answer

Small, sterile, bland vegetations along the valve closure line, associated with malignancy (especially mucinous adenocarcinoma) and hypercoagulability.

#113 · Cardiovascular and Vascular

Which organism usually causes acute infective endocarditis and in which valve in drug users?

Answer

Staphylococcus aureus, which destroys normal valves; the tricuspid valve is commonly affected in injecting drug users.

#114 · Cardiovascular and Vascular

Which organisms cause subacute endocarditis and on what valve?

Answer

Viridans streptococci on previously damaged valves.

#115 · Cardiovascular and Vascular

How do tricuspid endocarditis emboli appear on chest CT?

Answer

Multiple peripheral, nodular, often cavitating pulmonary septic emboli.

#116 · Cardiovascular and Vascular

What is the macroscopic appearance of fibrinous pericarditis?

Answer

A shaggy, 'bread and butter' appearance of the pericardial surfaces, seen in infarction, uraemia, radiation and rheumatic fever.

#117 · Cardiovascular and Vascular

What pericardial thickness suggests constrictive pericarditis on CT or MRI?

Answer

Thickening greater than about 4 mm, with or without calcification, plus dilated IVC and hepatic veins.

#118 · Cardiovascular and Vascular

Where do most cardiac myxomas arise and what is their attachment?

Answer

About 75 to 80 percent are in the left atrium, attached by a stalk to the interatrial septum at the fossa ovalis.

#119 · Cardiovascular and Vascular

What are the histology and syndrome associations of cardiac myxoma?

Answer

Stellate myxoma cells in an acid mucopolysaccharide matrix. Familial cases occur in Carney complex (PRKAR1A).

#120 · Cardiovascular and Vascular

Which cardiac tumour is commonest in children and with what disease?

Answer

Rhabdomyoma, associated with tuberous sclerosis, with 'spider cells' and frequent regression.

#121 · Cardiovascular and Vascular

What is the commonest primary malignant cardiac tumour in adults and where does it occur?

Answer

Angiosarcoma, usually in the right atrium.

#122 · Cardiovascular and Vascular

What are the four components of tetralogy of Fallot?

Answer

Ventricular septal defect, overriding aorta, right ventricular outflow (infundibular) obstruction and right ventricular hypertrophy.

#123 · Cardiovascular and Vascular

What are the radiographic signs of coarctation of the aorta in older patients?

Answer

Rib notching of the inferior surface of the posterior third to ninth ribs and the 3 sign at the coarctation site.

#124 · Cardiovascular and Vascular

What defines Stanford type A and type B aortic dissection and the usual management?

Answer

Type A involves the ascending aorta and is managed surgically; type B begins distal to the left subclavian artery and is managed medically unless complicated.

#125 · Cardiovascular and Vascular

What is the histological hallmark of thoracic aortic aneurysm in Marfan syndrome?

Answer

Cystic medial degeneration with loss of elastic fibres and smooth muscle and mucoid pools; Marfan syndrome involves FBN1.

#126 · Cardiovascular and Vascular

What are the features of giant cell arteritis on histology?

Answer

Granulomatous inflammation with multinucleated giant cells and fragmentation of the internal elastic lamina, in patients over 50.

#127 · Respiratory and Mediastinum

Which atelectasis mechanism produces mediastinal shift towards the collapse?

Answer

Resorption (obstructive) atelectasis, after complete airway obstruction by mucus, foreign body or tumour. Compression atelectasis from effusion or tension pneumothorax pushes the mediastinum away.

#128 · Respiratory and Mediastinum

What causes patchy (micro) atelectasis?

Answer

Loss of surfactant, as in neonatal respiratory distress syndrome and acute respiratory distress syndrome.

#129 · Respiratory and Mediastinum

Which emphysema pattern is linked to smoking and predominates in the upper lobes?

Answer

Centriacinar (centrilobular) emphysema, which affects the respiratory bronchioles and spares the distal alveoli.

#130 · Respiratory and Mediastinum

Which emphysema pattern characterises alpha-1 antitrypsin deficiency and where is it worst?

Answer

Panacinar emphysema, affecting the whole acinus uniformly and worst in the lower lobes.

#131 · Respiratory and Mediastinum

Which emphysema type is the typical source of spontaneous pneumothorax in young adults?

Answer

Distal acinar (paraseptal) emphysema, with subpleural bullae larger than 1 cm.

#132 · Respiratory and Mediastinum

What is the pathogenetic basis of smoking related emphysema?

Answer

An imbalance between proteases such as neutrophil elastase and antiproteases, with oxidative injury.

#133 · Respiratory and Mediastinum

Which microscopic findings characterise atopic asthma?

Answer

Thickened basement membrane, smooth muscle hypertrophy, mucous plugging, eosinophils, Charcot-Leyden crystals and Curschmann spirals.

#134 · Respiratory and Mediastinum

How is bronchiectasis recognised on CT?

Answer

The bronchus is wider than its accompanying artery (signet ring sign), with lack of tapering and bronchial wall thickening.

#135 · Respiratory and Mediastinum

Which bronchiectasis pattern suggests allergic bronchopulmonary aspergillosis?

Answer

Central bronchiectasis with mucoid impaction, in contrast to upper lobe predominance in cystic fibrosis.

#136 · Respiratory and Mediastinum

What histological pattern defines idiopathic pulmonary fibrosis?

Answer

Usual interstitial pneumonia: patchy subpleural fibrosis of varied age with fibroblastic foci beside normal lung.

#137 · Respiratory and Mediastinum

Which CT features constitute a definite UIP pattern?

Answer

Basal and subpleural reticulation with traction bronchiectasis and honeycombing.

#138 · Respiratory and Mediastinum

What is the pathogenetic model of idiopathic pulmonary fibrosis?

Answer

Repeated alveolar epithelial injury with aberrant repair driven by transforming growth factor beta.

#139 · Respiratory and Mediastinum

How does nonspecific interstitial pneumonia differ histologically from UIP?

Answer

It has uniform temporal appearance, whereas UIP is temporally heterogeneous. CT shows basal ground-glass with subpleural sparing.

#140 · Respiratory and Mediastinum

What are Masson bodies and in which pattern do they occur?

Answer

Plugs of loose granulation tissue in alveolar ducts and alveoli, in organising pneumonia, which responds to corticosteroids.

#141 · Respiratory and Mediastinum

Which CT sign is associated with organising pneumonia?

Answer

The reversed halo sign, with peripheral or peribronchovascular consolidation.

#142 · Respiratory and Mediastinum

Which CT findings suggest hypersensitivity pneumonitis?

Answer

Ground-glass opacity, centrilobular nodules and mosaic attenuation with air trapping on expiratory images; histology shows poorly formed non-necrotising granulomas.

#143 · Respiratory and Mediastinum

What distinguishes progressive massive fibrosis in coal workers' pneumoconiosis?

Answer

Black fibrotic masses larger than 1 cm, usually upper lobe, migrating towards the hila on imaging.

#144 · Respiratory and Mediastinum

Which findings characterise silicosis?

Answer

Upper lobe hard nodules with whorled collagen and birefringent particles, hilar node eggshell calcification, and increased risk of tuberculosis.

#145 · Respiratory and Mediastinum

What are asbestos bodies?

Answer

Asbestos fibres coated with iron and protein, appearing golden-brown and beaded; found in asbestosis.

#146 · Respiratory and Mediastinum

How do pleural plaques appear and where do they spare?

Answer

Discrete, often calcified fibrous plaques of the parietal pleura, sparing the costophrenic angles and apices.

#147 · Respiratory and Mediastinum

Which asbestos related malignancy is not increased by smoking?

Answer

Mesothelioma. Lung carcinoma risk is multiplied by smoking plus asbestos.

#148 · Respiratory and Mediastinum

What is the histological hallmark of sarcoidosis?

Answer

Tightly packed non-necrotising granulomas with epithelioid cells and giant cells; Schaumann and asteroid bodies are not specific.

#149 · Respiratory and Mediastinum

What is the classic chest imaging pattern of sarcoidosis?

Answer

Symmetric bilateral hilar and right paratracheal lymphadenopathy, with perilymphatic micronodules in the upper and middle zones.

#150 · Respiratory and Mediastinum

What is Löfgren syndrome?

Answer

Erythema nodosum, bilateral hilar lymphadenopathy and arthritis, a form of acute sarcoidosis with a good prognosis.

#151 · Respiratory and Mediastinum

What are the four stages of lobar pneumonia?

Answer

Congestion, red hepatisation, grey hepatisation and resolution.

#152 · Respiratory and Mediastinum

How does bronchopneumonia differ morphologically from lobar pneumonia?

Answer

Patchy peribronchial suppurative consolidation, often bilateral and basal, rather than homogeneous lobar consolidation.

#153 · Respiratory and Mediastinum

What is the Ghon complex?

Answer

The Ghon focus (subpleural primary focus) plus enlarged hilar nodes in primary tuberculosis; calcification gives the Ranke complex.

#154 · Respiratory and Mediastinum

Why does secondary tuberculosis favour the upper lobe apices?

Answer

High oxygen tension suits the aerobic organism; cavitation is frequent in a sensitised host.

#155 · Respiratory and Mediastinum

What is the microscopic appearance of tuberculous granuloma?

Answer

Caseating necrosis with epithelioid histiocytes, Langhans giant cells and a lymphocyte rim; organisms are shown by Ziehl-Neelsen stain.

#156 · Respiratory and Mediastinum

What imaging sign indicates endobronchial spread of tuberculosis?

Answer

Tree-in-bud nodules; miliary spread gives innumerable 1 to 3 mm randomly distributed nodules.

#157 · Respiratory and Mediastinum

What is the source of more than 95 percent of pulmonary emboli?

Answer

Deep vein thrombosis of the lower limbs.

#158 · Respiratory and Mediastinum

Why do most pulmonary emboli not cause infarction?

Answer

Dual supply from bronchial arteries; infarction usually needs pre-existing cardiac or lung disease and gives a wedge-shaped haemorrhagic subpleural infarct.

#159 · Respiratory and Mediastinum

What CT pulmonary angiography finding suggests right heart strain?

Answer

Enlarged right ventricle with bowing of the septum towards the left.

#160 · Respiratory and Mediastinum

What is the definition of pulmonary hypertension and its main heritable gene?

Answer

Mean pulmonary arterial pressure above 20 mmHg at rest; BMPR2 is the principal gene in heritable disease.

#161 · Respiratory and Mediastinum

What is the precursor sequence of lung adenocarcinoma?

Answer

Atypical adenomatous hyperplasia, adenocarcinoma in situ (3 cm or less, lepidic), minimally invasive adenocarcinoma (invasion 5 mm or less), then invasive adenocarcinoma.

#162 · Respiratory and Mediastinum

Which histological features and paraneoplastic effect belong to squamous cell carcinoma?

Answer

Central location, keratin pearls and intercellular bridges; PTHrP secretion causing hypercalcaemia.

#163 · Respiratory and Mediastinum

Which features define small cell lung carcinoma?

Answer

Small cells with nuclear moulding, crush artefact, high mitoses; positive for synaptophysin, chromogranin and CD56; ectopic ACTH, SIADH, Lambert-Eaton syndrome.

#164 · Respiratory and Mediastinum

How are typical and atypical carcinoid distinguished?

Answer

Typical: fewer than 2 mitoses per 2 mm squared and no necrosis. Atypical: 2 to 10 mitoses or punctate necrosis.

#165 · Respiratory and Mediastinum

Which immunohistochemical markers favour mesothelioma over adenocarcinoma?

Answer

Positive calretinin, WT1 and CK5/6; negative TTF-1 and CEA. BAP1 loss is typical.

#166 · Respiratory and Mediastinum

What are the CT features of malignant mesothelioma?

Answer

Nodular circumferential pleural thickening involving the mediastinal pleura, unilateral effusion and a contracted hemithorax.

#167 · Respiratory and Mediastinum

Which mediastinal tumour is associated with myasthenia gravis and how is it staged?

Answer

Thymoma, staged by Masaoka-Koga, with stage I encapsulated and stage IV pleural or pericardial dissemination or distant metastases.

#168 · Respiratory and Mediastinum

Which serum markers help distinguish mediastinal germ cell tumours?

Answer

Yolk sac tumour secretes alpha-fetoprotein; seminoma may mildly raise beta-hCG. Mature teratoma shows fat and calcification on CT.

#169 · Gastrointestinal and Hepatobiliary

What defines Barrett oesophagus histologically?

Answer

Replacement of distal squamous epithelium by columnar epithelium with intestinal metaplasia, recognised by goblet cells. It predisposes to oesophageal adenocarcinoma.

#170 · Gastrointestinal and Hepatobiliary

Which oesophageal cancer type predominates in the distal third and why?

Answer

Adenocarcinoma, driven by Barrett oesophagus, reflux and obesity. Squamous cell carcinoma favours the upper and middle thirds with tobacco and alcohol.

#171 · Gastrointestinal and Hepatobiliary

What is the pathological basis and barium appearance of primary achalasia?

Answer

Loss of myenteric plexus ganglion cells with failure of lower oesophageal sphincter relaxation. Barium shows a dilated oesophagus tapering to a beak.

#172 · Gastrointestinal and Hepatobiliary

What are the diagnostic histology and imaging signs of eosinophilic oesophagitis?

Answer

At least 15 intraepithelial eosinophils per high power field. Imaging shows fixed concentric rings (feline oesophagus) and a long narrow segment.

#173 · Gastrointestinal and Hepatobiliary

Which organism drives antral predominant gastritis and duodenal ulcer?

Answer

Helicobacter pylori. Pangastritis with atrophy and intestinal metaplasia instead leads to gastric ulcer and carcinoma.

#174 · Gastrointestinal and Hepatobiliary

Which features separate a malignant from a benign gastric ulcer on contrast study?

Answer

Malignant ulcers lie within a mass with a nodular, shouldered margin and clubbed, fused or amputated folds. Benign ulcers project beyond the gastric contour with smooth radiating folds.

#175 · Gastrointestinal and Hepatobiliary

How does the Lauren diffuse type of gastric cancer differ from the intestinal type?

Answer

Diffuse type is discohesive signet ring cells linked to CDH1 loss and produces linitis plastica. Intestinal type forms glands and follows atrophic gastritis and metaplasia.

#176 · Gastrointestinal and Hepatobiliary

How can gastric MALT lymphoma be distinguished from linitis plastica on imaging?

Answer

MALT lymphoma thickens folds or the wall but the stomach remains distensible. Linitis plastica gives a rigid, contracted, non-distensible stomach.

#177 · Gastrointestinal and Hepatobiliary

What are the immunohistochemical markers and driver mutations of a GIST?

Answer

KIT (CD117) and DOG1 positive, with activating KIT or PDGFRA mutations. It arises from interstitial cells of Cajal and is treated with imatinib.

#178 · Gastrointestinal and Hepatobiliary

What are the serological and histological hallmarks of coeliac disease?

Answer

Tissue transglutaminase antibodies with HLA-DQ2 or DQ8. Histology shows villous atrophy, crypt hyperplasia and raised intraepithelial lymphocytes.

#179 · Gastrointestinal and Hepatobiliary

Which malignant complications complicate coeliac disease?

Answer

Enteropathy associated T cell lymphoma and small bowel adenocarcinoma. Ulcerative jejunoileitis and hyposplenism are other complications.

#180 · Gastrointestinal and Hepatobiliary

What is the imaging fold pattern change in coeliac disease?

Answer

Jejunoileal fold pattern reversal, with fewer jejunal folds and more ileal folds. Transient non-obstructing intussusception is also common.

#181 · Gastrointestinal and Hepatobiliary

What are the pathological features of Crohn disease?

Answer

Transmural inflammation with skip lesions, aphthous then linear ulcers, cobblestone mucosa, creeping fat and fistulae. Non-caseating granulomas are found in a minority of biopsies.

#182 · Gastrointestinal and Hepatobiliary

Which imaging signs reflect Crohn disease in the small bowel?

Answer

Mural thickening with mucosal hyperenhancement, the comb sign of engorged vasa recta, and the string sign of a narrowed terminal ileum.

#183 · Gastrointestinal and Hepatobiliary

How does ulcerative colitis differ from Crohn disease in distribution and depth?

Answer

Ulcerative colitis is continuous from the rectum and confined to mucosa and superficial submucosa. Crohn disease is patchy, transmural and can affect any gut segment.

#184 · Gastrointestinal and Hepatobiliary

What imaging findings occur in chronic ulcerative colitis and toxic megacolon?

Answer

Loss of haustra with a featureless lead pipe colon and widened presacral space. Toxic megacolon is a transverse colon above 6 cm.

#185 · Gastrointestinal and Hepatobiliary

Why does small bowel neuroendocrine tumour cause carcinoid syndrome only with liver metastases?

Answer

Hepatic metastases release serotonin beyond first pass hepatic metabolism. The primary is usually distal ileal with a desmoplastic mesenteric mass.

#186 · Gastrointestinal and Hepatobiliary

Which diverticulum contains ectopic gastric mucosa and how is it detected?

Answer

Meckel diverticulum, a true diverticulum from the vitellointestinal duct. Technetium pertechnetate scintigraphy shows uptake.

#187 · Gastrointestinal and Hepatobiliary

What is the classic molecular sequence of the adenoma to carcinoma pathway?

Answer

APC loss, then KRAS activation, then TP53 loss. The serrated pathway instead involves BRAF mutation and methylation with microsatellite instability.

#188 · Gastrointestinal and Hepatobiliary

Which germline defects underlie Lynch syndrome and FAP?

Answer

Lynch syndrome is a germline mismatch repair gene mutation. Familial adenomatous polyposis is a germline APC mutation.

#189 · Gastrointestinal and Hepatobiliary

Which adenoma features carry higher malignant potential?

Answer

Villous architecture, size over 1 cm and high grade dysplasia. Hyperplastic polyps are small, left sided and not premalignant.

#190 · Gastrointestinal and Hepatobiliary

How is colorectal cancer T staged?

Answer

T1 submucosa, T2 muscularis propria, T3 through the muscularis propria into pericolorectal tissue, T4a visceral peritoneum, T4b adjacent organs.

#191 · Gastrointestinal and Hepatobiliary

Which features does staging rectal MRI report?

Answer

Distance of tumour from the mesorectal fascia, extramural vascular invasion and nodal features. A margin of 1 mm or less is a threatened fascia.

#192 · Gastrointestinal and Hepatobiliary

What is a colonic diverticulum pathologically and where is it commonest?

Answer

An acquired pseudodiverticulum, a mucosal herniation through the muscle wall where vasa recta penetrate. It is commonest in the sigmoid colon.

#193 · Gastrointestinal and Hepatobiliary

What imaging signs support acute appendicitis?

Answer

Non-compressible appendix wider than 6 mm with wall thickening and periappendiceal fat stranding, sometimes with an appendicolith.

#194 · Gastrointestinal and Hepatobiliary

What is the source of pseudomyxoma peritonei?

Answer

Gelatinous mucinous ascites usually from a ruptured low grade appendiceal mucinous neoplasm. It scallops visceral surfaces.

#195 · Gastrointestinal and Hepatobiliary

What does histology of chronic hepatitis B show that identifies HBsAg?

Answer

Ground glass hepatocytes. Interface hepatitis and lobular inflammation also occur, and HBV raises hepatocellular carcinoma risk even without cirrhosis.

#196 · Gastrointestinal and Hepatobiliary

How does a steatotic liver appear on CT, ultrasound and MRI?

Answer

Hypoattenuating on CT, hyperechoic on ultrasound, and signal drop on opposed phase MRI. Focal sparing occurs near the gallbladder fossa.

#197 · Gastrointestinal and Hepatobiliary

What genotype and MRI finding characterise primary haemochromatosis?

Answer

HFE C282Y homozygosity. The liver is markedly low signal on gradient echo and T2 weighted MRI with the spleen spared.

#198 · Gastrointestinal and Hepatobiliary

What are the liver histology findings of alpha-1 antitrypsin deficiency?

Answer

Periportal hepatocyte globules that are periodic acid Schiff positive and diastase resistant (PiZZ). Panlobular basal emphysema occurs in the lung.

#199 · Gastrointestinal and Hepatobiliary

What imaging changes characterise cirrhosis?

Answer

Nodular contour, right lobe and medial segment atrophy, caudate and left lateral hypertrophy, widened fissures and splenomegaly.

#200 · Gastrointestinal and Hepatobiliary

Why is the caudate lobe spared in Budd-Chiari syndrome?

Answer

It drains directly to the inferior vena cava, so it hypertrophies when the hepatic veins are obstructed.

#201 · Gastrointestinal and Hepatobiliary

Which duct size and pattern characterise primary sclerosing cholangitis versus primary biliary cholangitis?

Answer

Primary sclerosing cholangitis affects larger ducts with onion skin fibrosis and beading, linked to ulcerative colitis. Primary biliary cholangitis destroys small ducts with anti-mitochondrial antibodies.

#202 · Gastrointestinal and Hepatobiliary

What imaging pattern allows non-invasive diagnosis of hepatocellular carcinoma in cirrhosis?

Answer

Arterial phase hyperenhancement with washout on portal venous or delayed phase, often with an enhancing capsule (LI-RADS).

#203 · Gastrointestinal and Hepatobiliary

How does fibrolamellar carcinoma differ from focal nodular hyperplasia?

Answer

Fibrolamellar carcinoma affects young patients without cirrhosis and has a large mass with a central T2 low signal, often calcified scar. In focal nodular hyperplasia the scar is T2 bright.

#204 · Gastrointestinal and Hepatobiliary

What is the enhancement pattern of a hepatic haemangioma?

Answer

Peripheral discontinuous nodular enhancement that fills in centripetally, with very bright signal on heavily T2 weighted images.

#205 · Gastrointestinal and Hepatobiliary

Which hepatic adenoma subtype carries the highest malignant risk and which is steatotic?

Answer

Beta-catenin activated has the highest malignant risk. HNF1A inactivated is steatotic and loses signal on opposed phase imaging.

#206 · Gastrointestinal and Hepatobiliary

What are the ultrasound and scintigraphic features of acute cholecystitis?

Answer

Stones, wall thickening above 3 mm, pericholecystic fluid and a positive sonographic Murphy sign. Hepatobiliary scintigraphy shows non-filling gallbladder.

#207 · Gastrointestinal and Hepatobiliary

What risk factors and classes define cholangiocarcinoma?

Answer

Classed as intrahepatic, perihilar (Klatskin) or distal. Risks include primary sclerosing cholangitis, liver flukes, choledochal cysts and hepatolithiasis.

#208 · Gastrointestinal and Hepatobiliary

What does the 2012 Atlanta classification say about collections in pancreatitis?

Answer

Within 4 weeks they are acute peripancreatic fluid or acute necrotic collections. After 4 weeks they become pseudocyst or walled off necrosis.

#209 · Gastrointestinal and Hepatobiliary

Which molecular changes and imaging signs characterise pancreatic ductal adenocarcinoma?

Answer

KRAS, CDKN2A, TP53 and SMAD4 alterations from PanIN. Imaging shows a hypoenhancing head mass with the double duct sign.

#210 · Gastrointestinal and Hepatobiliary

How do serous cystadenoma, mucinous cystic neoplasm and IPMN differ?

Answer

Serous is benign, microcystic with a central scar and VHL link. Mucinous occurs in women in the body or tail with ovarian stroma and no duct communication. IPMN communicates with the duct.

#211 · Genitourinary Adrenal and Endocrine

What are the light microscopy and electron microscopy findings in minimal change disease?

Answer

Glomeruli look normal on light microscopy with negative immunofluorescence. Electron microscopy shows diffuse effacement of podocyte foot processes.

#212 · Genitourinary Adrenal and Endocrine

Which antibody target is found in most primary membranous nephropathy?

Answer

The phospholipase A2 receptor on podocytes. Immune complexes form subepithelial deposits with basement membrane spikes on silver stain.

#213 · Genitourinary Adrenal and Endocrine

Which primary glomerulonephritis is commonest worldwide, and what does immunofluorescence show?

Answer

IgA nephropathy, with mesangial IgA deposits. It presents with macroscopic haematuria a day or two after a mucosal infection.

#214 · Genitourinary Adrenal and Endocrine

What are the electron microscopy and serology features of post-infectious glomerulonephritis?

Answer

Subepithelial humps on electron microscopy and low serum C3, following group A streptococcal infection, mostly in children.

#215 · Genitourinary Adrenal and Endocrine

How do anti-GBM disease and ANCA vasculitis differ on immunofluorescence?

Answer

Anti-GBM disease shows linear IgG along the basement membrane. ANCA-associated vasculitis is pauci-immune, with few or no deposits, and both cause crescents.

#216 · Genitourinary Adrenal and Endocrine

What is the Kimmelstiel-Wilson lesion and which disease causes it?

Answer

Nodular glomerulosclerosis of diabetic nephropathy, the commonest cause of end-stage kidney disease, with hyaline arteriolosclerosis of afferent and efferent arterioles.

#217 · Genitourinary Adrenal and Endocrine

Which gene is mutated in X-linked Alport syndrome and what is the basement membrane finding?

Answer

COL4A5 mutation, with a lamellated glomerular basement membrane, deafness and ocular defects.

#218 · Genitourinary Adrenal and Endocrine

What does ultrasound typically show in chronic glomerular disease compared with acute disease?

Answer

Chronic disease gives small smooth echogenic kidneys with lost corticomedullary differentiation. Acute disease gives normal or enlarged kidneys.

#219 · Genitourinary Adrenal and Endocrine

What urine finding characterises acute tubular necrosis?

Answer

Muddy brown granular casts, from sloughed tubular epithelium, with a polyuric recovery phase after the oliguric phase.

#220 · Genitourinary Adrenal and Endocrine

Which organism most often causes acute pyelonephritis, and what are its complications?

Answer

Escherichia coli. Complications are papillary necrosis, pyonephrosis, perinephric abscess and emphysematous pyelonephritis in diabetics.

#221 · Genitourinary Adrenal and Endocrine

Where do scars of chronic reflux pyelonephritis lie and how do they appear?

Answer

Coarse scars, often upper pole, lie over clubbed dilated calyces, unlike the smooth uniform thinning of ischaemic or glomerular disease.

#222 · Genitourinary Adrenal and Endocrine

What are the features of xanthogranulomatous pyelonephritis?

Answer

Proteus infection with a staghorn calculus in an obstructed kidney, lipid-laden foamy macrophages microscopically, and the bear paw sign on CT.

#223 · Genitourinary Adrenal and Endocrine

Which cells characterise acute interstitial nephritis and what is the usual cause?

Answer

Interstitial lymphocytes and eosinophils, usually a hypersensitivity reaction to drugs such as penicillins, NSAIDs and proton pump inhibitors.

#224 · Genitourinary Adrenal and Endocrine

Name four associations of renal papillary necrosis.

Answer

Diabetes, analgesic nephropathy, sickle cell disease, and obstruction or tuberculosis. Sloughed papillae give ring shadows or filling defects.

#225 · Genitourinary Adrenal and Endocrine

Which genes cause autosomal dominant polycystic kidney disease?

Answer

PKD1 on chromosome 16 in about 85 percent, and PKD2 on chromosome 4.

#226 · Genitourinary Adrenal and Endocrine

What are the extrarenal associations of autosomal dominant polycystic kidney disease?

Answer

Hepatic cysts, intracranial berry aneurysms, mitral valve prolapse and colonic diverticula.

#227 · Genitourinary Adrenal and Endocrine

What gene and hepatic lesion define autosomal recessive polycystic kidney disease?

Answer

PKHD1, with congenital hepatic fibrosis causing portal hypertension. Perinatal presentation includes Potter sequence and echogenic salt and pepper kidneys.

#228 · Genitourinary Adrenal and Endocrine

Which cystic renal condition carries an increased risk of renal cell carcinoma and in whom?

Answer

Acquired cystic disease in long-term dialysis patients, whose kidneys are small.

#229 · Genitourinary Adrenal and Endocrine

What is the typical cause, site and imaging pattern of fibromuscular dysplasia of the renal artery?

Answer

Medial fibroplasia in young women, in the mid to distal artery, giving the string of beads appearance.

#230 · Genitourinary Adrenal and Endocrine

What vascular lesion characterises malignant hypertension in the kidney?

Answer

Hyperplastic arteriolitis with onion-skin thickening and fibrinoid necrosis, in contrast to hyaline arteriolosclerosis in benign nephrosclerosis.

#231 · Genitourinary Adrenal and Endocrine

Which genetic change underlies clear cell renal carcinoma?

Answer

Loss of function of the VHL gene on chromosome 3p, sporadic or in von Hippel-Lindau disease.

#232 · Genitourinary Adrenal and Endocrine

How does clear cell renal carcinoma look macroscopically, and how does it enhance?

Answer

Golden-yellow with haemorrhage and necrosis, and it may extend as a tumour thrombus. It is hypervascular, with avid corticomedullary enhancement.

#233 · Genitourinary Adrenal and Endocrine

What genetic associations and enhancement pattern define papillary renal carcinoma?

Answer

Trisomy 7 and 17, and MET in type 1 hereditary disease. It is hypovascular with mild homogeneous enhancement.

#234 · Genitourinary Adrenal and Endocrine

Which renal tumour shows perinuclear halos and raisinoid nuclei, and with which syndrome?

Answer

Chromophobe carcinoma, associated with Birt-Hogg-Dube syndrome, with a better prognosis.

#235 · Genitourinary Adrenal and Endocrine

Describe the macroscopic and microscopic features of renal oncocytoma.

Answer

A mahogany brown mass with a central stellate scar, composed of eosinophilic cells rich in mitochondria. The scar is not reliable enough to avoid biopsy.

#236 · Genitourinary Adrenal and Endocrine

What are the components of angiomyolipoma and its main complication?

Answer

Fat, smooth muscle and thick-walled abnormal vessels. Lesions above 4 cm risk spontaneous haemorrhage (Wunderlich syndrome), and it is linked to tuberous sclerosis.

#237 · Genitourinary Adrenal and Endocrine

What is the histological pattern of Wilms tumour and its imaging sign?

Answer

Triphasic: blastemal, epithelial and stromal. A claw sign of parenchyma wraps the mass, unlike neuroblastoma which displaces the kidney.

#238 · Genitourinary Adrenal and Endocrine

Give the T1 and T2 size thresholds for renal cell carcinoma.

Answer

T1a up to 4 cm, T1b above 4 to 7 cm, T2a above 7 to 10 cm, T2b above 10 cm, all confined to the kidney.

#239 · Genitourinary Adrenal and Endocrine

What defines T3a, T3b, T3c and T4 renal cell carcinoma?

Answer

T3a is renal vein, perinephric fat or collecting system. T3b is vena cava below the diaphragm and T3c above it. T4 is beyond Gerota fascia, including ipsilateral adrenal.

#240 · Genitourinary Adrenal and Endocrine

How are Bosniak 2019 classes IIF, III and IV distinguished?

Answer

IIF has smooth minimal wall or septal thickening or a hyperdense cyst of 3 cm or more, and needs follow-up. III has enhancing thick walls or septa, and IV has an enhancing nodule.

#241 · Genitourinary Adrenal and Endocrine

Which stone type is radiolucent on plain radiographs, and in what urine?

Answer

Uric acid, in acidic urine, in gout and high cell turnover. It is visible on CT.

#242 · Genitourinary Adrenal and Endocrine

Which stone forms staghorn calculi, and what is the cause?

Answer

Struvite, in alkaline urine with urease-producing bacteria such as Proteus.

#243 · Genitourinary Adrenal and Endocrine

Name risk factors for urothelial carcinoma and for bladder squamous cell carcinoma.

Answer

Smoking, aromatic amines, cyclophosphamide and pelvic radiation for urothelial carcinoma. Schistosoma haematobium for squamous carcinoma.

#244 · Genitourinary Adrenal and Endocrine

What are the T categories of bladder cancer by depth of invasion?

Answer

Ta non-invasive papillary, Tis in situ, T1 lamina propria, T2 muscularis propria, T3 perivesical tissue, T4a prostate, uterus or vagina, T4b pelvic or abdominal wall.

#245 · Genitourinary Adrenal and Endocrine

Where does prostate cancer commonly arise compared with benign prostatic hyperplasia?

Answer

Cancer arises in the peripheral zone (about 70 percent), whereas hyperplasia arises in the transition zone.

#246 · Genitourinary Adrenal and Endocrine

Give the ISUP grade groups by Gleason score.

Answer

Group 1 is 3+3, group 2 is 3+4, group 3 is 4+3, group 4 is 8, and group 5 is 9 or 10.

#247 · Genitourinary Adrenal and Endocrine

How do T1a, T1b and T1c prostate cancer differ?

Answer

T1a and T1b are incidental at resection, 5 percent or less versus more than 5 percent of tissue. T1c is found on needle biopsy for raised PSA.

#248 · Genitourinary Adrenal and Endocrine

Which immunohistochemical pattern supports prostate adenocarcinoma?

Answer

Absence of basal cells, so p63 is negative within malignant glands, with AMACR commonly positive.

#249 · Genitourinary Adrenal and Endocrine

Where does the left testis drain compared with the right?

Answer

Left testis drains to left para-aortic nodes near the renal hilum, right testis to interaortocaval nodes.

#250 · Genitourinary Adrenal and Endocrine

What is the pT staging of testicular tumours?

Answer

pT1 limited to testis without lymphovascular invasion, pT2 lymphovascular invasion or tunica vaginalis, pT3 spermatic cord, pT4 scrotum.

#251 · Genitourinary Adrenal and Endocrine

What are the risk factors and precursor of penile squamous cell carcinoma?

Answer

HPV, lack of circumcision, phimosis and lichen sclerosus, with penile intraepithelial neoplasia as precursor.

#252 · Genitourinary Adrenal and Endocrine

Where do ureteric stones lodge?

Answer

At the pelviureteric junction, the pelvic brim where the ureter crosses the iliac vessels, and the vesicoureteric junction.

#253 · Genitourinary Adrenal and Endocrine

Which pituitary adenoma is commonest functioning, and what are its features in women and men?

Answer

Prolactinoma. Women present with amenorrhoea and galactorrhoea, men later with hypogonadism and often a macroadenoma.

#254 · Genitourinary Adrenal and Endocrine

What size separates a pituitary microadenoma from a macroadenoma?

Answer

Under 10 mm is a microadenoma and 10 mm or more is a macroadenoma. Macroadenomas can compress the optic chiasm.

#255 · Genitourinary Adrenal and Endocrine

Which visual field defect does a pituitary macroadenoma classically cause, and why?

Answer

Bitemporal hemianopia, because compression from below affects the crossing nasal retinal fibres in the chiasm.

#256 · Genitourinary Adrenal and Endocrine

How does a pituitary microadenoma behave on early dynamic contrast MRI?

Answer

It enhances less than normal pituitary tissue and appears relatively hypointense in the early phase.

#257 · Genitourinary Adrenal and Endocrine

What is the stalk effect in a sellar mass?

Answer

Compression of the pituitary stalk removes dopamine inhibition, causing mild hyperprolactinaemia from any sellar mass, not only a prolactinoma.

#258 · Genitourinary Adrenal and Endocrine

What is Sheehan syndrome?

Answer

Pituitary infarction after postpartum haemorrhage and hypotension, because the enlarged gland of pregnancy is vulnerable to ischaemia.

#259 · Genitourinary Adrenal and Endocrine

What antibody and mechanism cause Graves disease?

Answer

Autoantibodies stimulate the TSH receptor, causing diffuse hyperplasia and unregulated hormone release.

#260 · Genitourinary Adrenal and Endocrine

How does scintigraphy separate Graves disease from subacute (de Quervain) thyroiditis in a thyrotoxic patient?

Answer

Graves shows diffusely increased uptake. Subacute granulomatous thyroiditis shows very low uptake, with a painful gland and giant cell granulomas.

#261 · Genitourinary Adrenal and Endocrine

What are the key histological and neoplastic associations of Hashimoto thyroiditis?

Answer

Lymphoplasmacytic infiltrate with germinal centres and oncocytic (Hurthle) change. It carries an increased risk of thyroid lymphoma, usually marginal zone or diffuse large B cell.

#262 · Genitourinary Adrenal and Endocrine

What is Riedel thyroiditis and why does it matter radiologically?

Answer

A fibrosing disease in the IgG4 related spectrum that extends beyond the capsule into neck structures, producing a hard fixed gland that mimics carcinoma.

#263 · Genitourinary Adrenal and Endocrine

What is the commonest cause of thyroid enlargement, and its pathogenesis?

Answer

Multinodular goitre, from cycles of hyperplasia and involution, usually driven by relative iodine deficiency.

#264 · Genitourinary Adrenal and Endocrine

Which ultrasound features of a thyroid nodule raise suspicion for malignancy?

Answer

Solid hypoechoic nodule, irregular or spiculated margin, microcalcifications, taller than wide shape and extrathyroidal extension.

#265 · Genitourinary Adrenal and Endocrine

What are the molecular and nuclear hallmarks of papillary thyroid carcinoma?

Answer

BRAF V600E, RET/PTC and RAS alterations. Nuclei are optically clear (Orphan Annie eye) with grooves and pseudoinclusions, and psammoma bodies may be present.

#266 · Genitourinary Adrenal and Endocrine

How does papillary thyroid carcinoma spread, and how do the nodal metastases look?

Answer

By lymphatics to cervical nodes. Nodal metastases are often cystic, with calcification or punctate echogenic foci, and may be the presenting feature.

#267 · Genitourinary Adrenal and Endocrine

What defines follicular thyroid carcinoma and why can it not be diagnosed on FNA alone?

Answer

Capsular or vascular invasion must be shown on histology. It spreads by blood to bone and lung and carries RAS and PAX8-PPARG changes.

#268 · Genitourinary Adrenal and Endocrine

What is NIFTP?

Answer

Noninvasive follicular thyroid neoplasm with papillary-like nuclear features: an encapsulated tumour without invasion that behaves indolently.

#269 · Genitourinary Adrenal and Endocrine

From which cell does medullary thyroid carcinoma arise, and what is the pathognomonic microscopic feature?

Answer

Parafollicular C cells, secreting calcitonin. Tumour nests lie in an amyloid stroma.

#270 · Genitourinary Adrenal and Endocrine

Where in the thyroid does medullary carcinoma usually lie, and why?

Answer

In the upper two thirds of the lobes, where C cells are concentrated. Coarse calcification is common.

#271 · Genitourinary Adrenal and Endocrine

Which thyroid cancer has the worst prognosis and which gene is typically mutated?

Answer

Anaplastic carcinoma, in older patients, with TP53 mutation, rapid growth and invasion of the trachea and vessels.

#272 · Genitourinary Adrenal and Endocrine

What proportion of primary hyperparathyroidism is due to adenoma, hyperplasia and carcinoma?

Answer

About 85 percent adenoma, 10 to 15 percent multigland hyperplasia and under 1 percent carcinoma.

#273 · Genitourinary Adrenal and Endocrine

What distinguishes parathyroid carcinoma from adenoma?

Answer

Invasion of surrounding tissue, vascular invasion or metastasis, not cytological atypia alone.

#274 · Genitourinary Adrenal and Endocrine

What is the most specific radiographic sign of hyperparathyroidism?

Answer

Subperiosteal bone resorption, best seen on the radial side of the middle phalanges of the index and middle fingers.

#275 · Genitourinary Adrenal and Endocrine

What is a brown tumour?

Answer

A lytic bone lesion of hyperparathyroidism composed of osteoclast rich fibrous tissue and haemosiderin; the skeletal syndrome is osteitis fibrosa cystica.

#276 · Genitourinary Adrenal and Endocrine

How do sestamibi scintigraphy and 4D CT localise a parathyroid adenoma?

Answer

Sestamibi shows persistent uptake as the thyroid washes out. 4D CT shows an arterially enhancing lesion with washout.

#277 · Genitourinary Adrenal and Endocrine

What is the rugger jersey spine and in which disease does it occur?

Answer

Alternating dense bands at vertebral end plates from osteosclerosis in renal osteodystrophy, due to secondary hyperparathyroidism.

#278 · Genitourinary Adrenal and Endocrine

What is the commonest cause of endogenous Cushing syndrome?

Answer

An ACTH secreting pituitary adenoma (Cushing disease). Other causes are adrenal adenoma or carcinoma and ectopic ACTH, classically from small cell lung cancer.

#279 · Genitourinary Adrenal and Endocrine

What biochemical pattern defines Conn syndrome?

Answer

Hypertension, hypokalaemia and low renin from autonomous aldosterone secretion, due to an adenoma or bilateral hyperplasia.

#280 · Genitourinary Adrenal and Endocrine

What unenhanced CT attenuation and MRI sign indicate a lipid rich adrenal adenoma?

Answer

Attenuation of 10 HU or less on unenhanced CT and signal loss on opposed phase chemical shift MRI.

#281 · Genitourinary Adrenal and Endocrine

What contrast washout thresholds characterise a lipid poor adrenal adenoma at 15 minutes?

Answer

Absolute washout of 60 percent or more, or relative washout of 40 percent or more.

#282 · Genitourinary Adrenal and Endocrine

What imaging features suggest adrenal cortical carcinoma?

Answer

Size over about 6 cm, heterogeneous enhancement, necrosis, calcification and venous invasion into the renal vein or inferior vena cava.

#283 · Genitourinary Adrenal and Endocrine

How is adrenal myelolipoma diagnosed on imaging?

Answer

Macroscopic fat on CT, with attenuation below minus 20 HU, within a benign tumour of fat and haematopoietic tissue.

#284 · Genitourinary Adrenal and Endocrine

What is the commonest cause of congenital adrenal hyperplasia?

Answer

21 hydroxylase deficiency, with cortisol deficiency, raised ACTH, androgen excess and bilateral thickened glands.

#285 · Genitourinary Adrenal and Endocrine

What adrenal appearance occurs in meningococcal Waterhouse-Friderichsen syndrome?

Answer

Bilateral adrenal haemorrhage, seen as enlarged high attenuation glands on CT.

#286 · Genitourinary Adrenal and Endocrine

Why is the rule of tens for phaeochromocytoma outdated?

Answer

Roughly 40 percent are hereditary (VHL, RET, NF1, SDHx), so the 10 percent figures understate genetic and bilateral disease.

#287 · Genitourinary Adrenal and Endocrine

How is malignancy defined in phaeochromocytoma or paraganglioma?

Answer

By metastases to sites without normal chromaffin tissue; histology cannot predict behaviour. SDHB mutation carries a higher metastatic risk.

#288 · Genitourinary Adrenal and Endocrine

What are the characteristic imaging features of phaeochromocytoma?

Answer

Heterogeneous enhancing mass above 10 HU unenhanced, often T2 bright (light bulb sign), imaged functionally with MIBG or 68Ga DOTATATE PET.

#289 · Genitourinary Adrenal and Endocrine

What are the WHO grades of pancreatic neuroendocrine tumour by Ki-67?

Answer

Grade 1 under 3 percent, grade 2 from 3 to 20 percent, grade 3 over 20 percent.

#290 · Genitourinary Adrenal and Endocrine

What is the typical CT enhancement of a well differentiated pancreatic neuroendocrine tumour?

Answer

Hypervascular, with avid arterial or pancreatic phase enhancement and often portal phase washout; liver metastases are also hypervascular.

#291 · Genitourinary Adrenal and Endocrine

Which functioning pancreatic neuroendocrine tumour is commonest, and what is its presentation?

Answer

Insulinoma, usually solitary, small and benign, causing fasting hypoglycaemia relieved by glucose (Whipple triad).

#292 · Genitourinary Adrenal and Endocrine

Which genes and tumour sets define MEN1, MEN2A and MEN2B?

Answer

MEN1 (menin): parathyroid, pancreatic and pituitary tumours. MEN2A (RET): medullary thyroid cancer, phaeochromocytoma, parathyroid hyperplasia. MEN2B (RET): medullary thyroid cancer, phaeochromocytoma, mucosal neuromas, marfanoid habitus, no parathyroid disease.

#293 · Genitourinary Adrenal and Endocrine

Which histological lesion is typical of diabetic nephropathy, and how does a diabetic foot present radiologically?

Answer

Nodular glomerulosclerosis (Kimmelstiel-Wilson nodules). In the foot, Charcot arthropathy affects Lisfranc and midfoot joints, and osteomyelitis lies adjacent to an ulcer over a pressure point.

#294 · Obstetric Gynaecological and Breast

Which HPV types cause most cervical cancer, and which cause condylomata?

Answer

High risk types 16 and 18 cause most cervical cancer (16 mainly squamous, 18 more adenocarcinoma). Low risk types 6 and 11 cause condylomata acuminata.

#295 · Obstetric Gynaecological and Breast

How do HPV E6 and E7 proteins drive cervical carcinogenesis?

Answer

E6 degrades p53 and E7 inactivates RB, removing cell cycle control in the transformation zone.

#296 · Obstetric Gynaecological and Breast

What is the histological difference between CIN 1 and CIN 3?

Answer

CIN 1 shows undifferentiated basaloid cells confined to the lower third of the epithelium, with maturation in the upper two thirds and koilocytes. CIN 3 shows undifferentiated basaloid cells through the full thickness.

#297 · Obstetric Gynaecological and Breast

What is the FIGO 2018 feature of stage IIIB cervical cancer?

Answer

Extension to the pelvic wall or hydronephrosis or a non functioning kidney. IIIC is pelvic or para-aortic nodal disease.

#298 · Obstetric Gynaecological and Breast

How does MRI show that parametrial invasion is absent in cervical cancer?

Answer

An intact low signal fibrous stromal ring around the intermediate to high T2 signal tumour excludes parametrial invasion.

#299 · Obstetric Gynaecological and Breast

What are the two pathways of vulvar squamous cell carcinoma?

Answer

HPV related (usually HPV 16, younger women, from high grade squamous intraepithelial lesion) and HPV independent (older women, lichen sclerosus background, TP53 mutation).

#300 · Obstetric Gynaecological and Breast

Which nodes are the first echelon for vulvar carcinoma?

Answer

The inguinofemoral nodes.

#301 · Obstetric Gynaecological and Breast

What defines extramammary Paget disease of the vulva?

Answer

An intraepidermal adenocarcinoma of large pale mucin containing cells, which may be associated with an underlying adnexal or visceral carcinoma.

#302 · Obstetric Gynaecological and Breast

What causes endometrial hyperplasia, and how does WHO classify it?

Answer

Unopposed oestrogen. WHO classes it as hyperplasia without atypia and atypical hyperplasia (endometrioid intraepithelial neoplasia), the latter with a substantial cancer risk.

#303 · Obstetric Gynaecological and Breast

Which genetic changes characterise endometrioid versus serous endometrial carcinoma?

Answer

Endometrioid: PTEN, PIK3CA, ARID1A, KRAS and microsatellite instability. Serous: TP53 mutation, arising in atrophic endometrium in older women, aggressive.

#304 · Obstetric Gynaecological and Breast

What are the four molecular groups of endometrial carcinoma and which has the best and worst prognosis?

Answer

POLE ultramutated (best), mismatch repair deficient, no specific molecular profile, and p53 abnormal (worst).

#305 · Obstetric Gynaecological and Breast

Which germline syndrome predisposes to endometrial cancer?

Answer

Lynch syndrome, from mismatch repair gene germline mutation.

#306 · Obstetric Gynaecological and Breast

What endometrial thickness in postmenopausal bleeding prompts biopsy, and how is myometrial invasion assessed?

Answer

Above 4 mm on transvaginal ultrasound. MRI assesses depth of invasion on dynamic contrast images, where a disrupted junctional zone suggests deep invasion.

#307 · Obstetric Gynaecological and Breast

What are the gross, microscopic and MRI features of a fibroid?

Answer

Gross: well circumscribed whorled white mass. Micro: fascicles of bland spindle cells, often MED12 mutated. MRI: low signal on T2 unless degenerated.

#308 · Obstetric Gynaecological and Breast

Which degeneration of a fibroid is typical in pregnancy?

Answer

Red (carneous) degeneration. Other forms are hyaline, cystic and calcific.

#309 · Obstetric Gynaecological and Breast

How is leiomyosarcoma distinguished from leiomyoma?

Answer

By coagulative necrosis, nuclear atypia and mitotic activity.

#310 · Obstetric Gynaecological and Breast

What is the MRI sign of adenomyosis?

Answer

A thickened junctional zone, over 12 mm, with small high signal foci. The uterus is globular and symmetrically enlarged.

#311 · Obstetric Gynaecological and Breast

What are the pathological and MRI features of an endometrioma?

Answer

Chocolate coloured fluid grossly. MRI shows high T1 signal that stays high with fat suppression and T2 shading.

#312 · Obstetric Gynaecological and Breast

Which ovarian cancers are associated with endometriosis?

Answer

Clear cell and endometrioid carcinomas.

#313 · Obstetric Gynaecological and Breast

What are the genetics and precursor of high grade serous ovarian carcinoma?

Answer

TP53 mutation in almost all cases, often with BRCA1 or BRCA2 germline mutation. The precursor is serous tubal intraepithelial carcinoma in the distal fallopian tube.

#314 · Obstetric Gynaecological and Breast

What should a bilateral mucinous ovarian tumour prompt?

Answer

A search for a metastatic primary, usually appendix or colon. Mucin in the peritoneum (pseudomyxoma peritonei) is usually appendiceal.

#315 · Obstetric Gynaecological and Breast

Which ovarian germ cell tumour is most common in young women, and what are its components?

Answer

Mature cystic teratoma (dermoid), with ectodermal, mesodermal and endodermal elements. Malignant change is rare and is squamous carcinoma.

#316 · Obstetric Gynaecological and Breast

Which ovarian germ cell tumour secretes alpha-fetoprotein and what is its histological hallmark?

Answer

Yolk sac tumour, with Schiller Duval bodies.

#317 · Obstetric Gynaecological and Breast

What are the hormone and histological features of granulosa cell tumour?

Answer

It secretes oestrogen, causing endometrial hyperplasia, and shows Call-Exner bodies.

#318 · Obstetric Gynaecological and Breast

What is Meigs syndrome?

Answer

Ovarian fibroma with ascites and a pleural effusion.

#319 · Obstetric Gynaecological and Breast

What is a Krukenberg tumour?

Answer

Bilateral ovarian metastases with signet ring cells, usually from the stomach.

#320 · Obstetric Gynaecological and Breast

What are the features of ovarian torsion on imaging?

Answer

An enlarged ovary with peripheral follicles and a twisted pedicle (whirlpool sign). Doppler flow may be reduced, though preserved flow does not exclude torsion.

#321 · Obstetric Gynaecological and Breast

What defines FIGO stage III ovarian cancer?

Answer

Peritoneal spread beyond the pelvis or retroperitoneal nodal metastasis. Stage IV is distant spread, including liver parenchymal metastasis or pleural effusion with positive cytology.

#322 · Obstetric Gynaecological and Breast

What is the commonest site of ectopic pregnancy and the key imaging finding?

Answer

The fallopian tube ampulla. Positive hCG with no intrauterine gestational sac, an adnexal ring and free fluid in the pouch of Douglas.

#323 · Obstetric Gynaecological and Breast

How do complete and partial hydatidiform moles differ in karyotype and p57?

Answer

Complete: diploid, usually 46,XX, all paternal, p57 negative, no fetus. Partial: triploid, usually 69,XXY, p57 positive, with some fetal parts.

#324 · Obstetric Gynaecological and Breast

What are the ultrasound features of a complete mole?

Answer

A snowstorm or cluster of grapes appearance of multiple cystic spaces without a fetus, often with ovarian theca lutein cysts.

#325 · Obstetric Gynaecological and Breast

What are the histology and first metastatic site of gestational choriocarcinoma?

Answer

Cytotrophoblast and syncytiotrophoblast proliferation with no villi, haemorrhage and necrosis. It spreads haematogenously, first to the lung, and responds well to chemotherapy.

#326 · Obstetric Gynaecological and Breast

How do accreta, increta and percreta differ?

Answer

Accreta: adherence to the myometrium. Increta: invasion into the myometrium. Percreta: invasion through the serosa to adjacent organs.

#327 · Obstetric Gynaecological and Breast

What imaging findings suggest placenta accreta?

Answer

Loss of the retroplacental clear zone, myometrial thinning and abnormal lacunae on ultrasound, and dark intraplacental bands on T2 MRI.

#328 · Obstetric Gynaecological and Breast

How do placenta praevia and abruption present differently?

Answer

Praevia: painless bleeding from placenta over the internal os. Abruption: painful bleeding with retroplacental haematoma and risk of disseminated intravascular coagulation.

#329 · Obstetric Gynaecological and Breast

What is the pathogenesis of pre-eclampsia?

Answer

Failed spiral artery remodelling by extravillous trophoblast causes placental ischaemia and release of antiangiogenic factors such as sFlt-1, producing endothelial dysfunction.

#330 · Obstetric Gynaecological and Breast

What does HELLP stand for and what neuroimaging complication accompanies eclampsia?

Answer

Haemolysis, elevated liver enzymes, low platelets. Posterior reversible encephalopathy syndrome shows bilateral posterior subcortical T2 hyperintensity.

#331 · Obstetric Gynaecological and Breast

How is breast cancer graded by the Nottingham system?

Answer

Tubule formation, nuclear pleomorphism and mitotic count, each scored 1 to 3. Total 3 to 5 is grade 1, 6 to 7 grade 2, and 8 to 9 grade 3.

#332 · Obstetric Gynaecological and Breast

What is the mammographic pattern of high grade DCIS and what is Paget disease of the nipple?

Answer

Clustered pleomorphic linear branching calcifications, with comedo necrosis histologically. Paget disease is DCIS cells spreading into the nipple epidermis.

#333 · Obstetric Gynaecological and Breast

What distinguishes invasive lobular from ductal carcinoma?

Answer

Discohesive single file cells from E-cadherin loss, frequently multifocal and bilateral, and subtle on mammography and ultrasound, so MRI is often used.

#334 · Obstetric Gynaecological and Breast

How do receptor groups determine breast cancer therapy and which is associated with BRCA1?

Answer

ER and PR positive: endocrine therapy. HER2 amplified: trastuzumab. Triple negative lacks all three and is more common in BRCA1 carriers.

#335 · Obstetric Gynaecological and Breast

What features of phyllodes tumour distinguish it from fibroadenoma?

Answer

Cellular stroma with leaf like clefts, rapid growth in a large lobulated mass, local recurrence, haematogenous rather than nodal spread, graded benign, borderline or malignant.

#336 · Musculoskeletal and Soft Tissue

What is the WHO definition of osteoporosis by DXA T-score?

Answer

A T-score of minus 2.5 or below. Bone mass is reduced with normal mineral to matrix ratio and trabecular architecture is lost.

#337 · Musculoskeletal and Soft Tissue

Which bone compartment is lost first in osteoporosis, and which sites fracture first?

Answer

Trabecular bone, because of its high surface area. Vertebral bodies, distal radius and femoral neck fracture first.

#338 · Musculoskeletal and Soft Tissue

How do osteomalacia and osteoporosis differ in pathology?

Answer

Osteomalacia is defective mineralisation of osteoid, giving widened osteoid seams. Osteoporosis has reduced but normally mineralised bone.

#339 · Musculoskeletal and Soft Tissue

What is the radiological sign of osteomalacia, and what does it represent?

Answer

Looser zones (pseudofractures), lucent bands perpendicular to the cortex. They are unmineralised osteoid stress lesions.

#340 · Musculoskeletal and Soft Tissue

What happens to the growth plate in rickets?

Answer

Cartilage fails to calcify, so the physis widens and the metaphysis becomes frayed, cupped and flared.

#341 · Musculoskeletal and Soft Tissue

What is osteitis fibrosa cystica?

Answer

The bone disease of hyperparathyroidism: osteoclast resorption with marrow fibrosis and brown tumours of giant cells and haemosiderin.

#342 · Musculoskeletal and Soft Tissue

Where is subperiosteal resorption most characteristic in hyperparathyroidism?

Answer

The radial side of the middle phalanges of the second and third fingers. Tuft resorption and a salt and pepper skull also occur.

#343 · Musculoskeletal and Soft Tissue

What spinal appearance characterises renal osteodystrophy?

Answer

The rugger jersey spine: sclerotic bands at the vertebral endplates alongside a less dense central body.

#344 · Musculoskeletal and Soft Tissue

What are the three phases of Paget disease of bone?

Answer

Lytic, mixed and sclerotic. Osteoclast overactivity comes first, followed by disorganised osteoblast bone formation.

#345 · Musculoskeletal and Soft Tissue

What is the histological hallmark of Paget disease?

Answer

A mosaic pattern of lamellar bone with prominent cement lines. Alkaline phosphatase is raised with normal calcium and phosphate.

#346 · Musculoskeletal and Soft Tissue

What malignant complication occurs in under 1 percent of Paget disease?

Answer

Osteosarcoma (sarcomatous transformation), which gives the second peak of osteosarcoma in older adults.

#347 · Musculoskeletal and Soft Tissue

What is the double line sign and what condition does it indicate?

Answer

On T2 weighted MRI, a bright inner granulation line and dark outer sclerotic rim. It indicates osteonecrosis.

#348 · Musculoskeletal and Soft Tissue

What is the histology of osteonecrosis?

Answer

Dead bone with empty lacunae and necrotic marrow, with a rim of reactive repair. Epiphyseal infarcts are wedge shaped and subchondral.

#349 · Musculoskeletal and Soft Tissue

Which organism is classically associated with osteomyelitis in sickle cell disease?

Answer

Salmonella, although Staphylococcus aureus remains the most common organism overall.

#350 · Musculoskeletal and Soft Tissue

Why does haematogenous osteomyelitis in children start in the metaphysis?

Answer

Slow flow in the sluggish venous sinusoids of the metaphysis lets bacteria settle.

#351 · Musculoskeletal and Soft Tissue

Define sequestrum, involucrum and cloaca.

Answer

Sequestrum is a fragment of dead bone. Involucrum is a sheath of reactive new bone around it. Cloaca is the channel through which pus drains.

#352 · Musculoskeletal and Soft Tissue

What is a Brodie abscess?

Answer

A subacute, well circumscribed intraosseous abscess, commonly in the metaphysis of a long bone.

#353 · Musculoskeletal and Soft Tissue

What does the penumbra sign indicate on MRI?

Answer

A T1 hyperintense rim of granulation tissue around an osteomyelitic abscess cavity.

#354 · Musculoskeletal and Soft Tissue

Which classic radiographic signs characterise osteosarcoma?

Answer

A Codman triangle and sunburst periosteal reaction with mixed lysis and sclerosis, usually in the metaphysis of the distal femur or proximal tibia.

#355 · Musculoskeletal and Soft Tissue

What cell and matrix define osteosarcoma?

Answer

A malignant spindle or pleomorphic cell tumour that directly produces osteoid. Spread is haematogenous, mainly to lung.

#356 · Musculoskeletal and Soft Tissue

Which genetic syndromes predispose to osteosarcoma?

Answer

Hereditary retinoblastoma (RB1) and Li-Fraumeni syndrome (TP53).

#357 · Musculoskeletal and Soft Tissue

What is the translocation and immunostain of Ewing sarcoma?

Answer

t(11;22) giving an EWSR1 and FLI1 fusion. The small round blue cells are CD99 positive.

#358 · Musculoskeletal and Soft Tissue

Describe the radiological appearance of Ewing sarcoma.

Answer

A permeative diaphyseal lesion with lamellated (onion skin) periosteal reaction and a large soft tissue mass, which mimics osteomyelitis.

#359 · Musculoskeletal and Soft Tissue

What is the most common primary malignant bone tumour of the adult pelvis and proximal femur?

Answer

Chondrosarcoma, which produces cartilage matrix, often with rings and arcs calcification.

#360 · Musculoskeletal and Soft Tissue

Which mutations are common in conventional chondrosarcoma?

Answer

IDH1 and IDH2 mutations.

#361 · Musculoskeletal and Soft Tissue

Where does giant cell tumour of bone arise and what are its neoplastic cells?

Answer

In the epiphysis extending to the subarticular surface after skeletal maturity. The mononuclear stromal cells are neoplastic; the giant cells are reactive.

#362 · Musculoskeletal and Soft Tissue

What genetic change is characteristic of giant cell tumour of bone?

Answer

H3F3A mutation (G34W). Denosumab, which blocks RANKL, is used in unresectable disease.

#363 · Musculoskeletal and Soft Tissue

Why does osteoid osteoma cause night pain relieved by NSAIDs?

Answer

The nidus (under 2 cm) produces prostaglandins. On imaging it is a lucent nidus with reactive cortical sclerosis.

#364 · Musculoskeletal and Soft Tissue

Which genes are mutated in hereditary multiple exostoses?

Answer

EXT1 and EXT2. A cartilage cap over 2 cm in an adult osteochondroma suggests malignant change.

#365 · Musculoskeletal and Soft Tissue

What gene is mutated in fibrous dysplasia and what is the McCune-Albright triad?

Answer

GNAS. The triad is polyostotic fibrous dysplasia, café au lait skin spots and endocrine hyperfunction.

#366 · Musculoskeletal and Soft Tissue

What are the histological findings of Langerhans cell histiocytosis?

Answer

Cells with grooved nuclei, CD1a and langerin positivity, and Birbeck granules. In the skull it gives a punched out lytic lesion.

#367 · Musculoskeletal and Soft Tissue

Which primary tumours metastasising to bone are typically sclerotic, lytic or mixed?

Answer

Prostate is sclerotic. Kidney and thyroid are lytic and expansile. Breast is often mixed.

#368 · Musculoskeletal and Soft Tissue

What are the pathological features of osteoarthritis?

Answer

Cartilage fibrillation and loss, eburnation of subchondral bone, osteophytes and subchondral cysts. Rheumatoid-type pannus is absent.

#369 · Musculoskeletal and Soft Tissue

Which hand joints are characteristically involved in osteoarthritis versus rheumatoid arthritis?

Answer

Osteoarthritis: distal interphalangeal joints (Heberden nodes) and first carpometacarpal. Rheumatoid arthritis: metacarpophalangeal, proximal interphalangeal and wrist, sparing distal joints.

#370 · Musculoskeletal and Soft Tissue

What is pannus and which disease produces it?

Answer

Inflamed granulation tissue of hyperplastic synovium that erodes cartilage and bone in rheumatoid arthritis.

#371 · Musculoskeletal and Soft Tissue

Describe the crystals of gout and their optical behaviour.

Answer

Needle shaped monosodium urate crystals, negatively birefringent, yellow when parallel to the compensator axis.

#372 · Musculoskeletal and Soft Tissue

How do gout erosions appear on imaging?

Answer

Punched out erosions with overhanging edges away from the joint margin, with joint space preserved until late.

#373 · Musculoskeletal and Soft Tissue

Which crystals cause chondrocalcinosis, and which joints are typical?

Answer

Calcium pyrophosphate: rhomboid, weakly positively birefringent. Knee menisci, triangular fibrocartilage, and radiocarpal and second and third MCP joints.

#374 · Musculoskeletal and Soft Tissue

What gene association links the seronegative spondyloarthropathies?

Answer

HLA-B27, with absent rheumatoid factor. Ankylosing spondylitis begins with bilateral symmetric sacroiliitis and progresses to a bamboo spine.

#375 · Musculoskeletal and Soft Tissue

Which molecular marker separates atypical lipomatous tumour from lipoma?

Answer

MDM2 and CDK4 amplification in atypical lipomatous tumour (well differentiated liposarcoma).

#376 · Musculoskeletal and Soft Tissue

What translocation defines synovial sarcoma?

Answer

t(X;18) producing an SS18 and SSX fusion. It is not derived from synovium and often occurs near large joints in younger adults.

#377 · Musculoskeletal and Soft Tissue

How does the zonal pattern of myositis ossificans differ from osteosarcoma?

Answer

Myositis ossificans is most mature at the periphery and least mature centrally. Osteosarcoma is the reverse.

#378 · Neuropathology and Head and Neck

Which vascular territories are affected in global hypoperfusion infarction of the brain?

Answer

The arterial watershed zones, between the anterior and middle cerebral territories and between the middle and posterior cerebral territories.

#379 · Neuropathology and Head and Neck

What type of necrosis follows cerebral infarction, and what are the microscopic stages?

Answer

Liquefactive necrosis. Red neurons from 12 to 24 hours, neutrophils at 24 to 72 hours, macrophages from about 3 to 5 days, then gliosis and a cystic cavity.

#380 · Neuropathology and Head and Neck

Which MRI sequence shows acute infarction within minutes when CT is normal?

Answer

Diffusion weighted imaging, which shows restricted diffusion from cytotoxic oedema.

#381 · Neuropathology and Head and Neck

Where does hypertensive intracerebral haemorrhage typically occur, versus cerebral amyloid angiopathy?

Answer

Hypertension: basal ganglia, thalamus, pons and cerebellum. Amyloid angiopathy: lobar haemorrhage in older patients with peripheral microbleeds on susceptibility sequences.

#382 · Neuropathology and Head and Neck

What is the commonest cause and site of non-traumatic subarachnoid haemorrhage?

Answer

Rupture of a saccular (berry) aneurysm, most often at the anterior communicating artery complex of the anterior circulation.

#383 · Neuropathology and Head and Neck

When does vasospasm typically occur after aneurysmal subarachnoid haemorrhage?

Answer

From about day 4 to day 14, causing delayed ischaemia.

#384 · Neuropathology and Head and Neck

What is the empty delta sign and which condition does it indicate?

Answer

A filling defect of thrombus in the superior sagittal sinus on contrast CT, indicating cerebral venous sinus thrombosis.

#385 · Neuropathology and Head and Neck

Which vessel bleeds in an extradural haematoma and what is its shape and relation to sutures?

Answer

The middle meningeal artery. Biconvex (lentiform), does not cross sutures, may cross dural reflections.

#386 · Neuropathology and Head and Neck

How does a subdural haematoma differ from an extradural haematoma in source and shape?

Answer

Venous bleeding from bridging veins, crescentic, crosses sutures but not dural reflections such as the falx.

#387 · Neuropathology and Head and Neck

Where are cerebral contusions and diffuse axonal injury typically found?

Answer

Contusions: crests of gyri in the inferior frontal and anterior temporal lobes. Diffuse axonal injury: grey white junction, corpus callosum splenium and dorsolateral brainstem.

#388 · Neuropathology and Head and Neck

Which imaging sequence best shows diffuse axonal injury when CT is often normal?

Answer

Gradient echo or susceptibility weighted MRI, showing small haemorrhagic foci at the typical sites.

#389 · Neuropathology and Head and Neck

What does cerebral abscess look like on MRI and how is it separated from necrotic tumour?

Answer

Ring enhancing lesion with a thin capsule and central restricted diffusion; necrotic tumour usually lacks central restricted diffusion.

#390 · Neuropathology and Head and Neck

What is the distribution and microscopic hallmark of herpes simplex encephalitis?

Answer

Haemorrhagic necrosis of the medial temporal and inferior frontal lobes; Cowdry type A intranuclear inclusions.

#391 · Neuropathology and Head and Neck

Which opportunistic infection of HIV causes asymmetric subcortical white matter demyelination without mass effect?

Answer

Progressive multifocal leukoencephalopathy, from JC virus infection of oligodendrocytes.

#392 · Neuropathology and Head and Neck

What are the four stages of neurocysticercosis?

Answer

Vesicular, colloidal vesicular, granular nodular and calcified nodular, caused by Taenia solium larvae.

#393 · Neuropathology and Head and Neck

What are the neuropathological findings in Creutzfeldt-Jakob disease?

Answer

Spongiform vacuolation of the neuropil, neuronal loss and gliosis without inflammation; cortical ribboning on DWI.

#394 · Neuropathology and Head and Neck

What are the pathological and MRI features of a multiple sclerosis plaque?

Answer

Sharply demarcated perivenular demyelination with relative axonal preservation; ovoid periventricular lesions perpendicular to the ventricles (Dawson fingers), with open ring enhancement when active.

#395 · Neuropathology and Head and Neck

What imaging finding distinguishes neuromyelitis optica spectrum disorder from multiple sclerosis in the cord?

Answer

Longitudinally extensive transverse myelitis spanning three or more vertebral segments, with aquaporin 4 antibodies.

#396 · Neuropathology and Head and Neck

What causes osmotic demyelination syndrome and where does it classically occur?

Answer

Rapid correction of chronic hyponatraemia; central pons (central pontine myelinolysis), with neurons and axons preserved.

#397 · Neuropathology and Head and Neck

What are the microscopic hallmarks of Alzheimer disease and the imaging pattern?

Answer

Extracellular beta amyloid plaques and intraneuronal tau neurofibrillary tangles; medial temporal and hippocampal atrophy.

#398 · Neuropathology and Head and Neck

What is the pathology of Parkinson disease?

Answer

Loss of pigmented dopaminergic neurons in the substantia nigra pars compacta, with Lewy bodies composed of alpha synuclein.

#399 · Neuropathology and Head and Neck

What genetic abnormality causes Huntington disease and what is the radiological sign?

Answer

CAG repeat expansion in the HTT gene on chromosome 4; caudate atrophy with enlarged frontal horns.

#400 · Neuropathology and Head and Neck

Which brain structures are affected in Wernicke encephalopathy and what is the deficiency?

Answer

Mammillary bodies, medial thalami, periaqueductal grey and tectal plate; thiamine deficiency.

#401 · Neuropathology and Head and Neck

How does WHO CNS5 classify adult type diffuse gliomas?

Answer

Astrocytoma IDH mutant (grade 2 to 4), oligodendroglioma IDH mutant and 1p/19q codeleted (grade 2 or 3), and glioblastoma IDH wildtype (grade 4).

#402 · Neuropathology and Head and Neck

Which molecular features allow an IDH wildtype diffuse astrocytic glioma to be diagnosed as glioblastoma without necrosis?

Answer

TERT promoter mutation, EGFR amplification, or combined gain of chromosome 7 and loss of chromosome 10.

#403 · Neuropathology and Head and Neck

What are the microscopic and imaging features of glioblastoma?

Answer

Pleomorphic astrocytes with pseudopalisading necrosis and microvascular proliferation; thick irregular rim enhancing mass with necrotic centre and oedema, often crossing the corpus callosum.

#404 · Neuropathology and Head and Neck

What are the microscopic and imaging features of oligodendroglioma?

Answer

Round nuclei with perinuclear halos and a chicken wire capillary pattern; frontal location with frequent calcification.

#405 · Neuropathology and Head and Neck

Describe pilocytic astrocytoma: grade, site, genetics and imaging.

Answer

Grade 1; children, cerebellum or optic pathway; BRAF alterations such as KIAA1549-BRAF; Rosenthal fibres; cyst with enhancing mural nodule.

#406 · Neuropathology and Head and Neck

What is the typical site and microscopic appearance of medulloblastoma?

Answer

Fourth ventricle and vermis in children; small round blue cells with Homer Wright rosettes; molecular groups WNT, SHH, group 3 and group 4.

#407 · Neuropathology and Head and Neck

What are the imaging features of meningioma and its risk factors?

Answer

Extra-axial dural based, avidly enhancing mass with dural tail and calcification; prior cranial radiation and neurofibromatosis type 2.

#408 · Neuropathology and Head and Neck

Which tumours are linked to neurofibromatosis type 2, and where does vestibular schwannoma arise?

Answer

Bilateral vestibular schwannomas, meningiomas and ependymomas; vestibular schwannoma arises from the vestibular nerve in the internal auditory canal and cerebellopontine angle.

#409 · Neuropathology and Head and Neck

What is the commonest intracranial tumour in adults and where does it lie?

Answer

Metastasis, at the grey white junction, from lung, breast, melanoma, renal and colorectal primaries.

#410 · Neuropathology and Head and Neck

Which intramedullary spinal tumour is commonest in adults and which in children?

Answer

Adults: ependymoma. Children: astrocytoma.

#411 · Neuropathology and Head and Neck

Where does myxopapillary ependymoma occur?

Answer

At the conus medullaris and filum terminale.

#412 · Neuropathology and Head and Neck

What are the features of schwannoma versus neurofibroma?

Answer

Schwannoma: encapsulated, Antoni A and B areas, Verocay bodies, S100 positive, target sign.
Neurofibroma: infiltrates the nerve; plexiform type is pathognomonic of NF1.

#413 · Neuropathology and Head and Neck

Which salivary gland tumour is most common, and what is its malignant complication?

Answer

Pleomorphic adenoma, mostly parotid; carcinoma ex pleomorphic adenoma.

#414 · Neuropathology and Head and Neck

What are the features of Warthin tumour?

Answer

Older male smokers, parotid tail, may be bilateral or multiple, oncocytic epithelium with lymphoid stroma, uptake on technetium pertechnetate.

#415 · Neuropathology and Head and Neck

Which malignant salivary tumour spreads along nerves, and what is its pattern?

Answer

Adenoid cystic carcinoma, with a cribriform pattern and perineural spread along nerves such as the facial and trigeminal.

#416 · Neuropathology and Head and Neck

What virus is associated with nasopharyngeal carcinoma and where does it arise?

Answer

Epstein-Barr virus; the fossa of Rosenmuller, with cervical nodal metastases.

#417 · Neuropathology and Head and Neck

What is cholesteatoma and where does it erode?

Answer

Accumulation of keratinising squamous epithelium in the middle ear, often from pars flaccida retraction, eroding the scutum and ossicles.

#418 · Neuropathology and Head and Neck

What are the features of retinoblastoma?

Answer

RB1 loss in young children, leukocoria, intraocular calcification.

#419 · Neuropathology and Head and Neck

Which extraocular muscle is most commonly involved in Graves orbitopathy and what is spared?

Answer

The inferior rectus; muscle bellies enlarge and the tendons are spared.

#420 · Haematology Paediatric and Infection

Which blood film cells and gastric pathology define pernicious anaemia?

Answer

Macro-ovalocytes and hypersegmented neutrophils from B12 deficiency, due to autoimmune atrophic gastritis of the body and fundus with intrinsic factor antibodies. It carries an increased risk of gastric adenocarcinoma and neuroendocrine tumour.

#421 · Haematology Paediatric and Infection

What are the imaging findings of beta thalassaemia major in the skull and chest?

Answer

Marrow expansion gives a hair-on-end skull, widened ribs and coarse trabeculae. Extramedullary haematopoiesis forms paravertebral soft tissue masses.

#422 · Haematology Paediatric and Infection

How is cardiac iron overload quantified in thalassaemia, and what does a low value mean?

Answer

MRI T2* of the myocardium. Shortened (low) T2* indicates heavy iron loading and predicts heart failure.

#423 · Haematology Paediatric and Infection

What molecular change causes sickle cell disease?

Answer

A glutamic acid to valine substitution at position 6 of the beta globin chain (HbS). Deoxygenation polymerises haemoglobin and cells sickle.

#424 · Haematology Paediatric and Infection

What are the skeletal imaging features of sickle cell disease?

Answer

Bone infarcts, avascular necrosis of the femoral head and H-shaped (step-off) vertebral endplates. Salmonella osteomyelitis is a recognised complication.

#425 · Haematology Paediatric and Infection

What happens to the spleen in an adult with sickle cell disease and why?

Answer

Repeated infarction causes autosplenectomy, leaving a small fibrotic, often calcified spleen. Children may instead have splenomegaly.

#426 · Haematology Paediatric and Infection

Which marrow appearance defines aplastic anaemia on pathology and MRI?

Answer

A hypocellular, fat-replaced marrow without fibrosis or infiltrate, seen on MRI as diffuse high T1 marrow signal.

#427 · Haematology Paediatric and Infection

What is the defining genetic abnormality of chronic myeloid leukaemia?

Answer

t(9;22), the Philadelphia chromosome, producing the BCR::ABL1 fusion with constitutive tyrosine kinase activity. It is treated with tyrosine kinase inhibitors such as imatinib.

#428 · Haematology Paediatric and Infection

Which mutation is found in most polycythaemia vera and what thrombotic complications occur?

Answer

JAK2 V617F. Arterial and venous thrombosis occur, including Budd-Chiari syndrome and portal or mesenteric vein thrombosis.

#429 · Haematology Paediatric and Infection

What are the pathological and imaging features of primary myelofibrosis?

Answer

Marrow fibrosis, teardrop cells on a leukoerythroblastic film and splenomegaly from extramedullary haematopoiesis. Imaging shows osteosclerosis, low T1 and T2 marrow signal and a very large spleen.

#430 · Haematology Paediatric and Infection

What translocation defines acute promyelocytic leukaemia and how does it present and respond?

Answer

t(15;17) producing PML::RARA. It presents with disseminated intravascular coagulation and responds to all-trans retinoic acid.

#431 · Haematology Paediatric and Infection

What is a myeloid sarcoma and where does it occur?

Answer

An extramedullary tumour mass of myeloblasts (chloroma), seen as a soft tissue mass in the orbit, spine or skin, and it may precede overt leukaemia.

#432 · Haematology Paediatric and Infection

Which cytogenetic findings predict good and poor prognosis in childhood ALL?

Answer

t(12;21) ETV6::RUNX1 is favourable, as is age 2 to 10 years. t(9;22) is poor. Sanctuary sites are the central nervous system and testis.

#433 · Haematology Paediatric and Infection

What immunophenotype and film feature characterise CLL, and what is Richter transformation?

Answer

Small mature B cells expressing CD5 and CD23, with smudge cells on the film. Richter transformation is evolution to diffuse large B-cell lymphoma, suggested by rapid node growth and high FDG avidity.

#434 · Haematology Paediatric and Infection

What is the diagnostic cell and immunophenotype of classic Hodgkin lymphoma?

Answer

The Reed-Sternberg cell, a large binucleate cell with prominent nucleoli, expressing CD30 and usually CD15 and typically CD20 negative.

#435 · Haematology Paediatric and Infection

Which Hodgkin lymphoma subtype is commonest and how does it differ from nodular lymphocyte predominant disease?

Answer

Nodular sclerosis is commonest, with fibrous bands and lacunar cells. Nodular lymphocyte predominant disease has popcorn cells that are CD20 positive and CD30 negative.

#436 · Haematology Paediatric and Infection

How does the pattern of spread differ between Hodgkin and most non-Hodgkin lymphoma?

Answer

Hodgkin lymphoma spreads to contiguous nodal groups. Most non-Hodgkin lymphoma spreads non-contiguously and more often involves extranodal sites.

#437 · Haematology Paediatric and Infection

What are the B symptoms of lymphoma?

Answer

Fever above 38 degrees Celsius, drenching night sweats and loss of more than 10 percent of body weight in 6 months.

#438 · Haematology Paediatric and Infection

What translocation defines follicular lymphoma and what does it do?

Answer

t(14;18) places BCL2 under the immunoglobulin heavy chain locus, so neoplastic follicles resist apoptosis. Grade 3B behaves like DLBCL.

#439 · Haematology Paediatric and Infection

What is the molecular hallmark of Burkitt lymphoma and its histological appearance?

Answer

MYC translocation, most often t(8;14). It shows a starry sky pattern with a Ki-67 index near 100 percent.

#440 · Haematology Paediatric and Infection

Which translocation and protein define mantle cell lymphoma?

Answer

t(11;14) causing cyclin D1 overexpression, with classical cases SOX11 positive. Lymphomatous polyposis is a gastrointestinal manifestation.

#441 · Haematology Paediatric and Infection

What is the association and regression behaviour of gastric MALT lymphoma?

Answer

It is associated with Helicobacter pylori and early disease may regress after eradication. t(11;18) API2::MALT1 predicts non-response.

#442 · Haematology Paediatric and Infection

How does primary CNS lymphoma appear on imaging in immunocompetent versus HIV patients?

Answer

Immunocompetent patients show solitary or multiple homogeneously enhancing periventricular masses with restricted diffusion. HIV-related EBV-driven tumours more often show ring enhancement.

#443 · Haematology Paediatric and Infection

Define the four Ann Arbor stages of lymphoma.

Answer

I is one nodal region, II is two or more regions on the same side of the diaphragm, III is nodal regions on both sides of the diaphragm, and IV is diffuse extranodal organ involvement.

#444 · Haematology Paediatric and Infection

What does the Lugano classification change about Ann Arbor staging?

Answer

It uses FDG PET-CT for avid lymphomas, groups I and II as limited and III and IV as advanced, and drops A and B designation for non-Hodgkin lymphoma.

#445 · Haematology Paediatric and Infection

How is PET response graded in lymphoma and which scores indicate complete metabolic response?

Answer

The Deauville five-point scale from 1 (no uptake) to 5 (uptake markedly higher than liver or new lesions). Scores 1 to 3 usually indicate complete metabolic response.

#446 · Haematology Paediatric and Infection

What diagnostic criteria define myeloma and what do the CRAB features stand for?

Answer

10 percent or more clonal plasma cells or a proven plasmacytoma with end-organ damage. CRAB is hypercalcaemia, renal impairment, anaemia and bone lesions.

#447 · Haematology Paediatric and Infection

What are the typical imaging findings of myeloma and why is bone scintigraphy poor?

Answer

Multiple punched-out lytic lesions without sclerotic rim and vertebral compression fractures. Scintigraphy underestimates disease because osteoblastic activity is low.

#448 · Haematology Paediatric and Infection

How does MGUS differ from myeloma and what is its progression risk?

Answer

Less than 10 percent marrow plasma cells, low paraprotein and no end-organ damage. It progresses to myeloma at about 1 percent per year.

#449 · Haematology Paediatric and Infection

What special stain and appearance confirm AL amyloid, and what is the cardiac MRI feature?

Answer

Congo red with apple-green birefringence under polarised light. Cardiac MRI shows thick ventricular walls with diffuse subendocardial late gadolinium enhancement.

#450 · Haematology Paediatric and Infection

Which mutation and paraprotein characterise Waldenstrom macroglobulinaemia?

Answer

MYD88 L265P with an IgM paraprotein causing hyperviscosity. Lytic bone lesions are uncommon.

#451 · Haematology Paediatric and Infection

What are the immunophenotype and ultrastructural marker of Langerhans cell histiocytosis?

Answer

Cells express CD1a, langerin (CD207) and S100, and electron microscopy shows Birbeck granules. BRAF V600E is often present.

#452 · Haematology Paediatric and Infection

What is the skeletal imaging appearance of Langerhans cell histiocytosis?

Answer

A lytic skull lesion with a bevelled edge and vertebra plana in the spine. Pituitary stalk involvement can cause diabetes insipidus.

#453 · Haematology Paediatric and Infection

How do Rosai-Dorfman disease and Erdheim-Chester disease differ in histology and imaging?

Answer

Rosai-Dorfman has emperipolesis, S100 positive and CD1a negative histiocytes and massive cervical nodes. Erdheim-Chester has CD68 positive CD1a negative cells, symmetrical long bone sclerosis, hairy kidney and coated aorta.

#454 · Haematology Paediatric and Infection

Which granulomatous pattern distinguishes tuberculous, sarcoid and cat scratch lymphadenitis?

Answer

Tuberculosis shows caseating necrosis, sarcoidosis shows non-caseating granulomas, and cat scratch disease (Bartonella henselae) shows necrotising stellate granulomas with neutrophils.

#455 · Haematology Paediatric and Infection

What are the features of unicentric versus multicentric Castleman disease?

Answer

The unicentric hyaline vascular form is a solitary avidly enhancing mass. Multicentric disease may be HHV8 related, particularly in HIV, with systemic inflammatory symptoms.

#456 · Haematology Paediatric and Infection

What is the commonest splenic cyst and what separates it from a true epidermoid cyst?

Answer

The post-traumatic pseudocyst is commonest and lacks an epithelial lining. A true epidermoid cyst has a squamous lining.

#457 · Haematology Paediatric and Infection

What are the commonest primary benign and overall malignant splenic tumours?

Answer

Haemangioma is the commonest primary benign tumour. Lymphoma is the commonest malignant splenic tumour overall.

#458 · Haematology Paediatric and Infection

What are the pathological and skeletal features of Gaucher disease?

Answer

Glucocerebrosidase deficiency with storage cells showing crumpled tissue paper cytoplasm, causing massive splenomegaly and an Erlenmeyer flask deformity of the distal femur.

#459 · Haematology Paediatric and Infection

What are the WHO thymoma types, staging system and syndromes associated with thymoma?

Answer

Types A, AB, B1, B2 and B3, with thymic carcinoma separate, staged by Masaoka-Koga. Associations are myasthenia gravis, pure red cell aplasia and hypogammaglobulinaemia.

#460 · Haematology Paediatric and Infection

How can MRI separate thymic hyperplasia from a thymic tumour?

Answer

Chemical shift imaging: hyperplasia loses signal on out-of-phase images, whereas a tumour does not.

#461 · Haematology Paediatric and Infection

What deficiency causes neonatal respiratory distress syndrome, and what is the histological hallmark?

Answer

Surfactant deficiency from immature type II pneumocytes. Alveoli are collapsed and lined by eosinophilic hyaline membranes of fibrin and cell debris.

#462 · Haematology Paediatric and Infection

What is the chest radiograph appearance of neonatal RDS?

Answer

Bilateral symmetrical fine granular (ground glass) opacification with air bronchograms and low lung volumes.

#463 · Haematology Paediatric and Infection

Which maternal factors raise or lower the risk of neonatal RDS?

Answer

Maternal diabetes raises risk by delaying surfactant synthesis. Antenatal corticosteroids lower risk by accelerating type II pneumocyte maturation.

#464 · Haematology Paediatric and Infection

What is bronchopulmonary dysplasia and what causes it?

Answer

Chronic lung disease of arrested alveolar and vascular development after prolonged oxygen and ventilation in the premature lung. Imaging shows coarse reticular opacity with hyperlucent cystic areas.

#465 · Haematology Paediatric and Infection

Which bowel segments are most often involved in necrotising enterocolitis?

Answer

The terminal ileum and right colon. Prematurity is the dominant risk factor.

#466 · Haematology Paediatric and Infection

What radiographic signs indicate necrotising enterocolitis and its perforation?

Answer

Pneumatosis intestinalis (gas in the bowel wall) and portal venous gas indicate NEC. Pneumoperitoneum indicates perforation.

#467 · Haematology Paediatric and Infection

What is the late complication of necrotising enterocolitis in the colon?

Answer

Stricture formation. Resection can also leave short bowel syndrome.

#468 · Haematology Paediatric and Infection

What is the characteristic brain lesion of hypoxic ischaemic injury in the preterm infant?

Answer

Periventricular leukomalacia, necrosis of white matter around the lateral ventricles, with cysts on later cranial ultrasound.

#469 · Haematology Paediatric and Infection

Which brain regions are damaged by a severe acute hypoxic ischaemic insult in the term infant?

Answer

Deep grey matter (thalami and basal ganglia), perirolandic cortex and brainstem. Early MRI shows restricted diffusion.

#470 · Haematology Paediatric and Infection

Where does germinal matrix haemorrhage arise and in whom?

Answer

From fragile capillaries in the subependymal germinal matrix of premature infants. It is graded by ventricular and parenchymal extension and is seen on ultrasound.

#471 · Haematology Paediatric and Infection

Distinguish malformation, disruption and deformation.

Answer

Malformation is an intrinsic developmental defect. Disruption is secondary destruction of normal tissue, such as amniotic bands. Deformation is abnormal mechanical force on normal tissue, such as from oligohydramnios.

#472 · Haematology Paediatric and Infection

What is the Potter sequence?

Answer

Renal agenesis causing oligohydramnios, which produces pulmonary hypoplasia and limb deformities.

#473 · Haematology Paediatric and Infection

Which gestational period carries the greatest teratogenic risk for structural malformation?

Answer

The embryonic period, weeks 3 to 9.

#474 · Haematology Paediatric and Infection

Which supplement lowers the risk of neural tube defects, and which maternal test is raised in them?

Answer

Folic acid lowers risk. Maternal serum and amniotic fluid alpha-fetoprotein are raised in open defects.

#475 · Haematology Paediatric and Infection

Which defect underlies congenital diaphragmatic hernia, and what causes death?

Answer

Failure of pleuroperitoneal canal closure, most often left sided through the foramen of Bochdalek. Death is from pulmonary hypoplasia.

#476 · Haematology Paediatric and Infection

What is the commonest form of oesophageal atresia?

Answer

A blind upper pouch with a fistula from the trachea to the distal oesophagus. A coiled nasogastric tube sits in the upper pouch with gas in the stomach.

#477 · Haematology Paediatric and Infection

Which four anatomical abnormalities define tetralogy of Fallot in a newborn with cyanosis?

Answer

Ventricular septal defect, pulmonary outflow obstruction, overriding aorta and right ventricular hypertrophy. The radiograph shows a boot-shaped heart with oligaemic lungs.

#478 · Haematology Paediatric and Infection

What is Eisenmenger syndrome?

Answer

Reversal of a left to right shunt to a right to left shunt after pulmonary hypertension develops from chronic overcirculation.

#479 · Haematology Paediatric and Infection

Which genetic change marks aggressive neuroblastoma?

Answer

MYCN amplification, with older age and undifferentiated histology also adverse.

#480 · Haematology Paediatric and Infection

What are the histological features of neuroblastoma?

Answer

Small round blue cells with Homer Wright rosettes and neuropil. Maturation gives ganglioneuroblastoma and ganglioneuroma.

#481 · Haematology Paediatric and Infection

Which investigations help diagnose neuroblastoma?

Answer

Raised urinary vanillylmandelic and homovanillic acid, and MIBG scintigraphy. Imaging shows calcification and vessel encasement rather than invasion.

#482 · Haematology Paediatric and Infection

What is stage 4S neuroblastoma?

Answer

Metastatic disease in infants limited to skin, liver and bone marrow with a small primary, with a good prognosis and frequent spontaneous regression.

#483 · Haematology Paediatric and Infection

Which syndromes are associated with Wilms tumour?

Answer

WAGR and Denys-Drash (WT1), and Beckwith-Wiedemann syndrome (11p15 locus).

#484 · Haematology Paediatric and Infection

What is the triphasic histology of Wilms tumour?

Answer

Blastemal, epithelial (tubular) and stromal elements. Anaplasia worsens prognosis.

#485 · Haematology Paediatric and Infection

How does Wilms tumour differ from neuroblastoma on imaging?

Answer

Wilms is an intrarenal mass with a claw sign, displaces vessels, rarely calcifies and may extend into the renal vein and IVC. Neuroblastoma often calcifies and encases vessels.

#486 · Haematology Paediatric and Infection

What is the commonest primary liver malignancy of childhood and which marker is raised?

Answer

Hepatoblastoma, occurring under 3 years, with raised serum alpha-fetoprotein.

#487 · Haematology Paediatric and Infection

Which syndromes predispose to hepatoblastoma?

Answer

Beckwith-Wiedemann syndrome and familial adenomatous polyposis, with prematurity also a risk factor.

#488 · Haematology Paediatric and Infection

What is the botryoid variant of embryonal rhabdomyosarcoma?

Answer

A grape-like polypoid mass in the vagina or bladder. Tumour cells are desmin and myogenin positive.

#489 · Haematology Paediatric and Infection

Which translocation characterises alveolar rhabdomyosarcoma?

Answer

PAX3 or PAX7 fusion to FOXO1, usually in adolescents with limb tumours.

#490 · Haematology Paediatric and Infection

Which gene is lost in retinoblastoma and what is trilateral disease?

Answer

RB1 loss. Trilateral disease is bilateral retinoblastoma with a pineoblastoma, and survivors carry a risk of later osteosarcoma.

#491 · Haematology Paediatric and Infection

What are the histological and imaging features of retinoblastoma?

Answer

Small round blue cells with Flexner-Wintersteiner rosettes, necrosis and calcification. Imaging shows a calcified intraocular mass, and the presenting sign is leukocoria.

#492 · Haematology Paediatric and Infection

What accumulates in Gaucher disease and what are its bone and organ findings?

Answer

Glucocerebroside from glucocerebrosidase deficiency. Findings are lipid-laden macrophages, hepatosplenomegaly and an Erlenmeyer flask deformity of the distal femur.

#493 · Haematology Paediatric and Infection

What enzyme defect causes Tay-Sachs disease?

Answer

Hexosaminidase A deficiency with ganglioside accumulation in neurons, giving a cherry-red macular spot.

#494 · Haematology Paediatric and Infection

Which glycogen storage disease is a lysosomal disorder with cardiomegaly?

Answer

Pompe disease (type II), acid maltase deficiency. Von Gierke disease (type I) instead gives hepatomegaly, hypoglycaemia and hepatic adenomas.

#495 · Haematology Paediatric and Infection

What are the pulmonary imaging findings of cystic fibrosis?

Answer

Upper lobe predominant bronchiectasis with bronchial wall thickening, from CFTR mutation (commonly F508del).

#496 · Haematology Paediatric and Infection

What is the usual lung pattern of lobar pneumonia and its stages?

Answer

Whole lobe consolidation passing through congestion, red hepatisation, grey hepatisation and resolution. Imaging shows consolidation with air bronchograms.

#497 · Haematology Paediatric and Infection

What are the Ghon focus, Ghon complex and Ranke complex?

Answer

The Ghon focus is the subpleural primary lesion. With hilar nodal involvement it is the Ghon complex, and when both calcify it is the Ranke complex.

#498 · Haematology Paediatric and Infection

What is the histology of a tuberculous granuloma?

Answer

Caseating necrosis surrounded by epithelioid macrophages, Langhans giant cells and lymphocytes.

#499 · Haematology Paediatric and Infection

What are the typical imaging findings of primary tuberculosis in a child compared with reactivation in an adult?

Answer

Children show hilar and mediastinal lymphadenopathy. Adult reactivation shows apical cavitating disease.

#500 · Haematology Paediatric and Infection

How do cerebral toxoplasmosis and primary CNS lymphoma differ in HIV?

Answer

Toxoplasmosis gives multiple ring enhancing lesions, often in the basal ganglia. Primary CNS lymphoma is usually a solitary EBV associated lesion.